Conditioning and Non-Convexity of Semantic Decoding by Inversion of Deep Brain Encoding Models

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Abhranil Sarkar (23BML0077) · Tanisha Choudhuri (23BML0081)

Guide: Dr. Sharmila Nageswaran, Associate Professor, School of Electronics Engineering

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Motivation: reading meaning back out of the brain

Clinical need

ALS, brainstem stroke, locked-in syndrome and severe aphasia can leave comprehension intact while removing speech and writing. A non-invasive decoder would restore a channel without neurosurgery.

Encoders are now very good

Deep encoding models predict whole-cortex fMRI from video, audio and language, and accuracy keeps rising with model size and data [3, 7]. Decoding is usually assumed to follow.

The untested step

A forward model can be highly predictive and still have an inverse that is non-unique or unstable (Hadamard; Hansen [8]). Nobody had measured this for a deep foundation encoder.

Question. When an encoder becomes very accurate, does it become invertible? We answer it for Meta's open 177 M-parameter TRIBE v2.

Project at a glance: complete

28.4×

lower encoding error, while recovery saturates at 0.944

4.8×

error barrier between distinct equally good blind solutions, 4 truths (8.1× to a deeper one)

−0.125

visual cortex, most responsive, recovers worst vs random vertices

4 / 5

held-out real-fMRI segments identified (chance 2.2%), cosine 0.02

Completion against the project plan



Status: 100% complete. All six objectives met. Delivered: IEEE-format manuscript, final report (VIT 2026 format), A3 poster, open GitHub repository with code, logged results, an interactive live demo and four Colab notebooks. Key claims re-verified on the real model on 19 October 2026.

Problem, gap and objectives

Gap. Encoders keep improving [3, 7], but nobody has measured whether that makes meaning more recoverable from fMRI, or what limits the inverse. Linear encoders are invertible only up to a linear transform [18]; deep encoders were never tested.

01	Formulate decoding as inversion; find the differentiable entry point	Done
02	Synthetic framework with exact ground truth	Done
03	Characterise the inverse vs. budget, noise, extent, priors	Done
04	Discriminate non-identifiability vs. non-convexity vs. optimiser limits	Done
05	Linear baseline; recoverability by anatomical network	Done
06	Validate on real naturalistic fMRI (LeBel 2023)	Done

Literature review

Work	Contribution	Limitation relevant here
Mitchell et al., 2008	Linear model predicts fMRI for unseen nouns	Single words, hand-built features
Kay et al., 2008 [21]	Identifies viewed images via predicted responses	Identification, not recovery
Naselaris et al., 2011 [1]	Encoding models as the route to decoding	Assumes invertibility
Huth et al., 2016 [2]	Semantic maps across cortex from natural speech	Linear; no inverse analysis
Antonello et al., 2023 [3]	Encoding accuracy scales with model size	Scaling shown for encoding only
Tang et al., 2023 [4]	Continuous language from fMRI via LM search	Per-subject decoder; search, not inversion
Gardner & Liu, 2019 [18]	Linear IEMs identifiable only up to a transform	Linear models only
Lévy et al., 2025 [6]	Typed sentences from MEG (CER 32%) and EEG (67%)	Shows EEG's limits for language
d'Ascoli et al., 2026, TRIBE v2 [7]	Open foundation encoder, 720 subjects	Encoder only; inverse never studied
Bladon & Bent; Benchetrit et al., 2026 [19, 20]	Feature visualisation; synthetic training data	Neither tests uniqueness of the inverse

Objectives anchored to the literature

Prior work	What it assumed or left open	What we add
Naselaris et al. 2011 [1]	Encoding models are the route to decoding	Measure when that route fails: dissociation of 28× vs +0.015
Antonello et al. 2023 [3]; TRIBE v2 2026 [7]	Encoding accuracy scales log-linearly with data	Show decoding saturates while encoding error keeps falling
Tang et al. 2023 [4]	Decoding via LM-guided search, per-subject decoder	Decoder-free inversion; exact error under ground truth
Hansen 2010 [8] (inverse problems)	Ill-posedness via conditioning and null spaces	Test both on a 177 M-parameter operator: non-convex, not null-space
Huth 2016 [2]; Fedorenko 2024 [10]; Buckner 2008 [11]	Semantic, language and default-mode networks	Recoverability and modality sensitivity per network
Brain2Qwerty 2026 [6]; Défossez 2023 [22]	Non-invasive decoding limited by modality	Limits that hold even for a near-perfect fMRI forward model

Proof of novelty I: prior-art comparison

Closest prior work	Nonlinear deep encoder	Optimises encoder input	Tests uniqueness / conditioning	Region-wise recoverability
Kay et al., Nature 2008 [21]	○	○	○	○
Gardner & Liu, eNeuro 2019 [18]	○	●	●	○
Tang et al., Nat. Neurosci. 2023 [4]	○	●	○	●
d'Ascoli et al., TRIBE v2, 2026 [7]	●	○	○	○
Bladon & Bent, arXiv May 2026 [19]	●	●	○	○
Benchetrit et al., arXiv Jun 2026 [20]	●	○	○	○
This work	●	●	●	●

Only this work fills all four columns. The closest result, Gardner & Liu [18], proves that a linear inverted encoding model recovers the stimulus only up to an invertible linear transform. We ask the same question of a 177 M-parameter nonlinear encoder, where no closed-form answer exists, and answer it empirically with exact ground truth.

Proof of novelty II: search protocol and the closest TRIBE v2 work

Search protocol

Date: 9 October 2026

Sources: arXiv, Google Scholar, general web

Queries: "TRIBE v2"; invert / inversion / identifiability / conditioning / non-convex + "brain encoding model" / "fMRI encoder"

Inclusion: any work that inverts, optimises the input of, or analyses the identifiability of a brain encoding model

Result: two TRIBE v2 follow-ups (right). Neither tests whether the inverse is unique or how it is conditioned.

Bladon & Bent, May 2026 [19]

What it does: Feature visualisation: optimises inputs to maximise a region's predicted response and recovers known cortical selectivity.

Difference: Asks what a region prefers. Does not ask whether the input can be recovered, or whether the answer is unique.

Benchetrit et al., June 2026 [20]

What it does: Uses TRIBE v2 forwards to synthesise fMRI that augments training of a brain-to-image decoder.

Difference: The encoder is only run forwards; its inverse is never analysed.

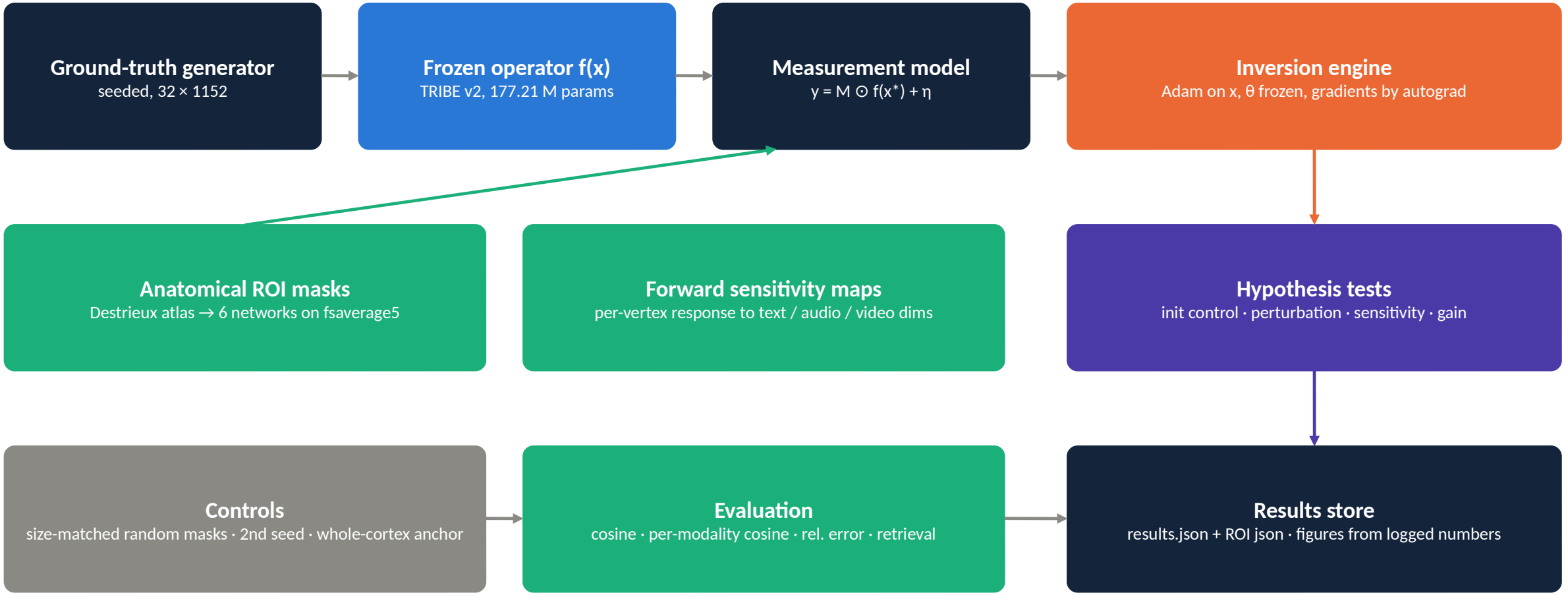
Novel contributions, each with its evidence

#	Contribution	Evidence	Status
1	First uniqueness and conditioning analysis of a deep foundation brain encoder	Prior-art matrix and dated search protocol (previous two slides)	Shown
2	Encoding accuracy and decodability dissociate	Encoding error falls 28.4×; the final halving buys +0.015 recovery	Shown
3	Incomplete recovery is non-convexity, not a null space	Equal fits disagree (min cosine 0.817 ± 0.007, 4 truths); straight paths between distinct ones rise 4.8× (2.4–8.2×), 8.1× to a deeper minimum	Shown
4	Error concentrates in the operator's least-sensitive directions	Solution difference 2.15 ± 0.17× below the random median over 4 truths (1.9× originally)	Shown
5	Predictable is not recoverable, region by region	Visual −0.125, somatomotor +0.102 vs size-matched random sets; ≈10× the control spread	Shown
6	Nonlinear inversion beats linear decoding	0.918 vs 0.060 for the best ridge decoder (≈15×)	Shown
7	On real fMRI the inverse identifies but does not reconstruct	Held-out story: 4/5 segments among 45 (chance 2.2%), cosine 0.02; oracle 5/5, 0.22	Shown

Methodology, analytical model and validation

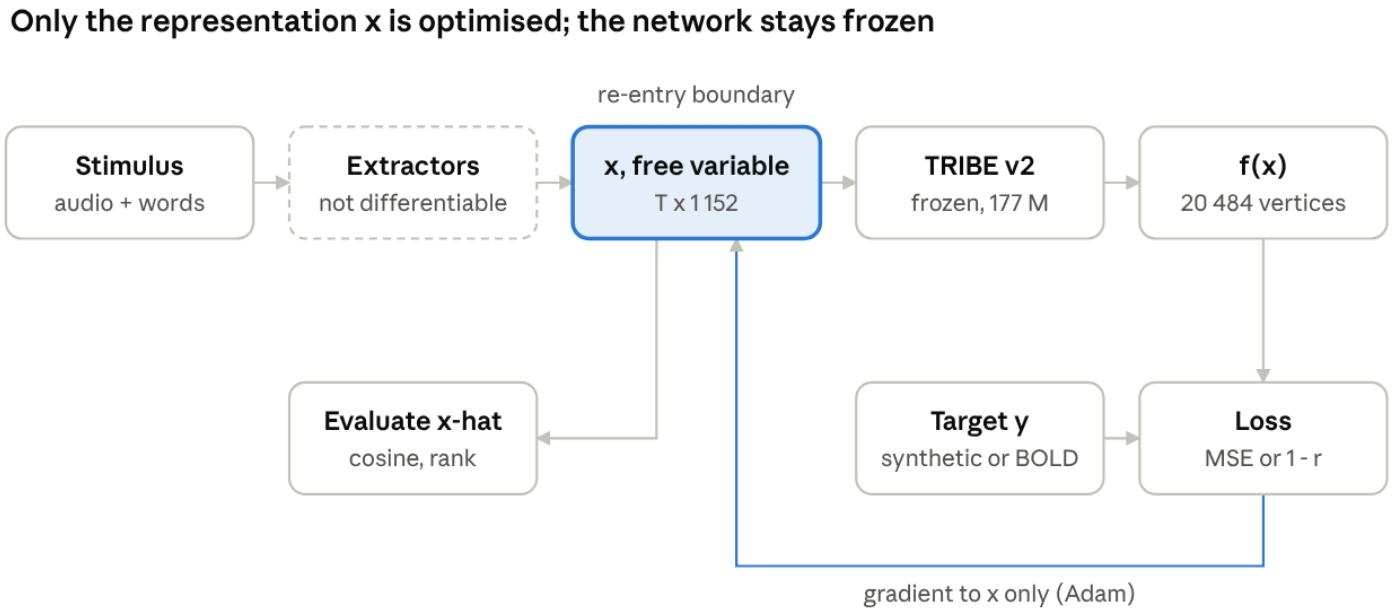
How every claim is tested, and what would have falsified it

Functional block diagram (final)



New since Review II: ROI masks, sensitivity maps, hypothesis tests and controls (green / violet / grey).

Where we cut into the network



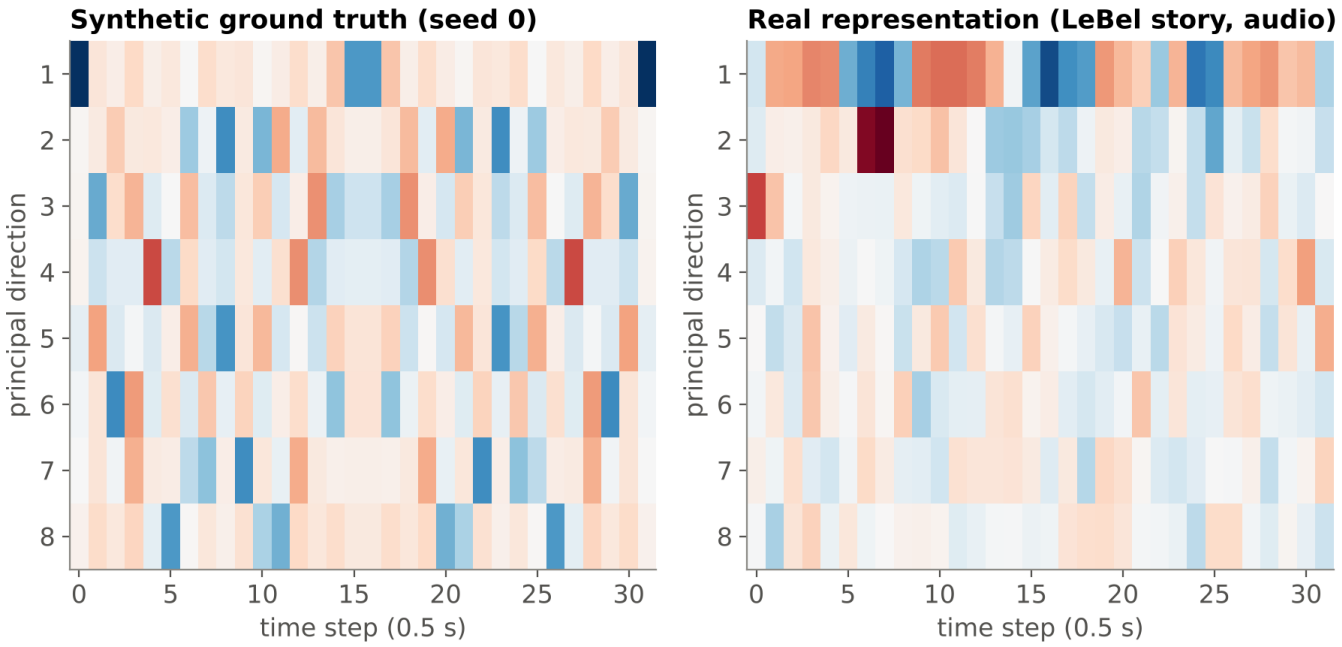
Fact	Value
Parameters (all frozen)	177.21 M: encoder 127.47 M, low-rank head 1,152 → 2,048, subject layer 41.97 M, temporal pooler
Input we optimise	x ∈ ℝ^(T × 1,152): text | audio | video slots of 384 each, 2 Hz
Output	20,484 fsaverage5 vertices × 100 steps at 1 Hz
Re-entry check	reproduces the released predictions exactly (r = 1.0000); $\|\partial f/\partial x\| = 7.97 \times 10^{3}$, no zero entries

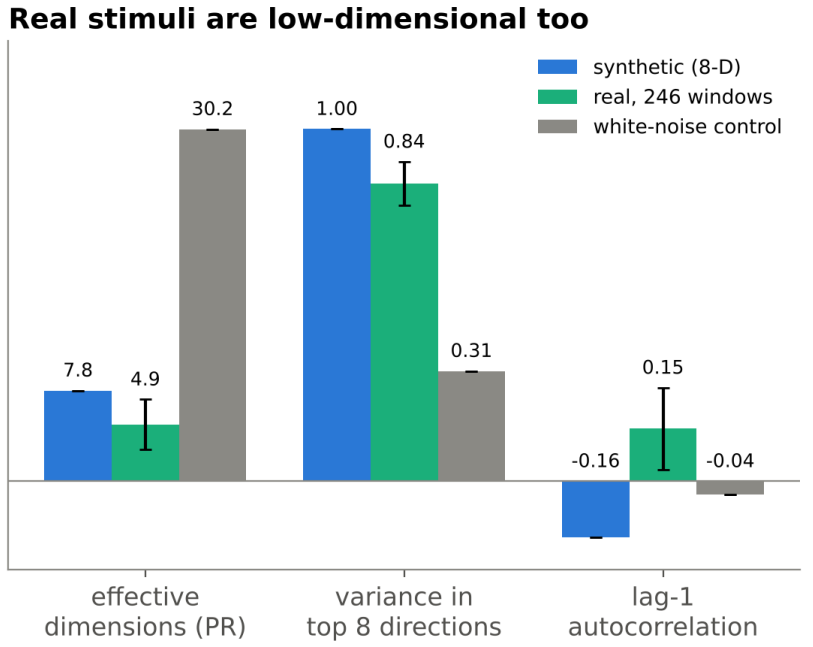
Inversion and the synthetic protocol

$$\hat{x} = \operatorname{argmin}_x \|f(x) - y\|^2 + \lambda R(x) \quad \cdot \quad s(v) = \|f(x + \varepsilon v) - f(x)\| / \varepsilon$$

Setting	Value
Optimiser	Adam [17], lr 5×10^{-2}, cosine annealing over the budget; θ frozen, gradients by autograd
Initialisation	0.01·N(0, I), seed 999 (default); eight distributions for the uniqueness study
Ground truth	8 random directions in ℝ^1152, each with its own cosine time course (2–16 cycles per 16 s), T = 32, unit s.d.; seeds 0–3; white noise as off-manifold control
Target	y = f(x), so recovery error is exact; real data: 1 − Pearson r over time (scale-free)
Metrics	mean per-step cosine to x, relative error, encoding error, retrieval rank among time steps
Priors tested	ℓ_2 and temporal smoothness, λ swept; none by default

The synthetic ground truth: how it is built and why it is valid

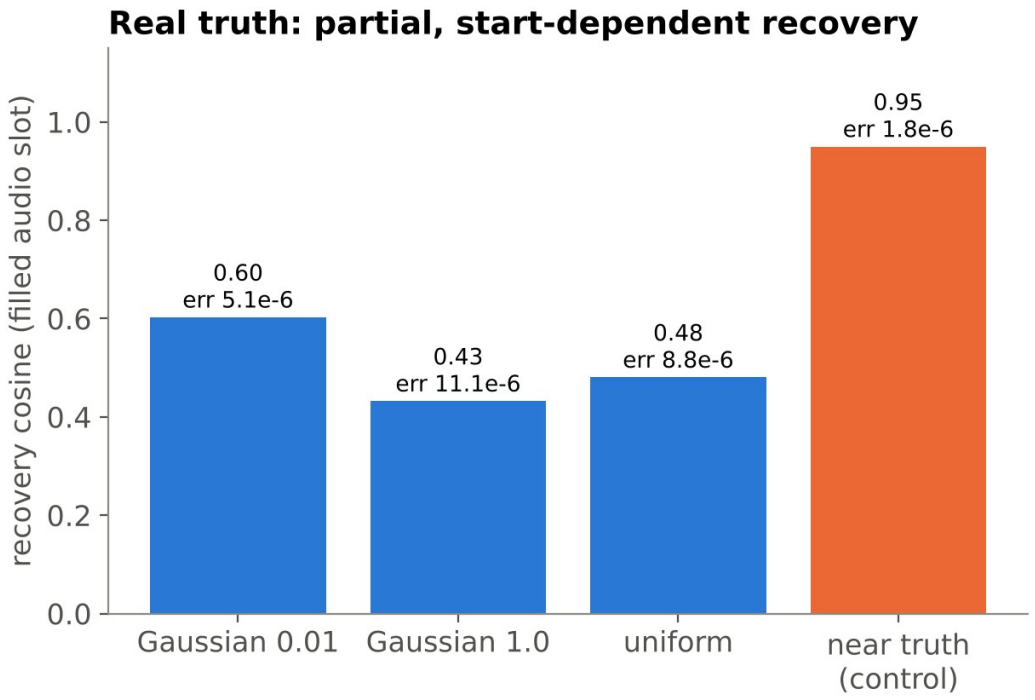


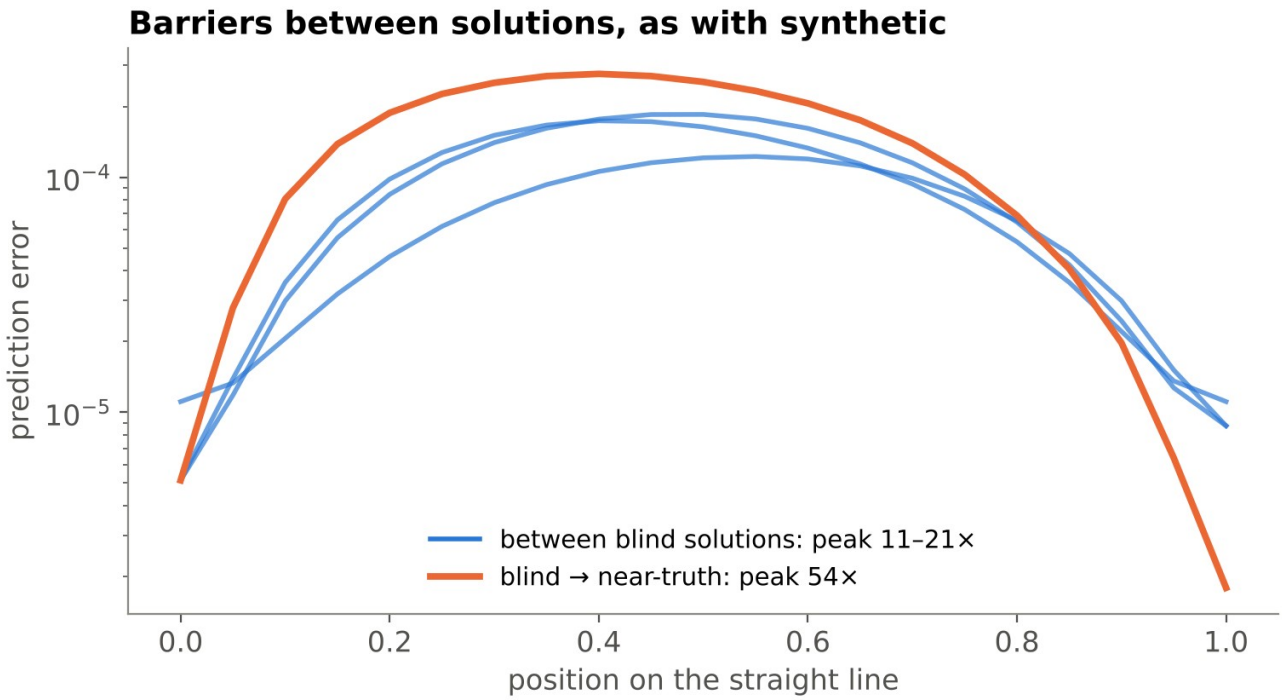


Construction. x lives in TRIBE v2's own input space (32 steps × 1,152, text | audio | video): 8 random directions, each with its own cosine time course (2–16 cycles per 16 s), scaled to unit s.d.; four seeds. y = f(x) through the frozen model, so the recovery error is exact by construction (re-entry r = 1.0000).

Representative of real stimuli. Real LeBel representations: 4.9 ± 2.1 effective dimensions, 84% of variance in 8 directions, s.d. 1.10 (synthetic 7.8, 100%, 1.00). White noise recovers the same (0.947 vs 0.942); a 64× change in speed, spanning real smoothness, changes nothing (0.836–0.858).

Check: a real stimulus representation as ground truth





Property	Synthetic ground truth	Real ground truth (wheretheressmoke, 16 s)
Recovery from blind starts	≈ 0.56 at 3,000 steps (0.94 at 32,000)	0.43–0.60 at 3,000 steps (audio slot)
Agreement between blind starts	0.89–1.00	0.19–0.56
Barrier between blind solutions	median 4.8× (2.4–8.2×)	11–21×
Deeper minimum near the truth	4.2× lower error, 8.1× barrier	2.9× lower error, 54× barrier

What TRIBE v2 learned from: its fMRI training data

Dataset (train)	What participants did in the scanner	Subjects	Sessions	fMRI (h)	TR (s)
CNeuroMod / Algonauts 2025 [23]	Watched TV (Friends, seasons 1–6) and 4 feature films: video + audio + dialogue	4	1,408	268.7	1.49
BOLD Moments [24]	Watched 1,102 three-second naturalistic video clips (1,000 train, 102 test × 10)	10	520	61.9	1.75
LeBel et al. 2023 [5]	Listened to spoken autobiographical stories (podcasts); 82 stories for 3 subjects	8	432	85.8	2.0
Wen et al. 2018 [25]	Watched natural movie clips (8-min segments, test segments ×10)	3	258	35.2	2.0
Training total	All 3T; passive viewing / listening, no task	**25**	**2,618**	**451.6**	—

Evaluation (never trained on). NNDb (86 subjects, films), Le Petit Prince (112, audiobook), Narratives (321, spoken stories), HCP 7T (176, movie clips): 695 subjects, 666.1 h. With training: 720 subjects, 5,094 sessions, 1,117.7 h of fMRI [7].

What this means for us. Our real-data subject UTS03 is one of the 8 LeBel training subjects, but our test story (wheretheressmoke) is TRIBE v2's own held-out test split, so the encoder never saw the responses we invert. The released weights use one averaged 'unseen subject' layer, so we invert the population model.

Why fMRI, not EEG

Criterion	fMRI (this work)	EEG
Spatial detail	≈2–3 mm voxels; 20,484 cortical vertices	32–128 scalp channels, cm-scale, blurred
Foundation encoder to invert	TRIBE v2: 177 M parameters, 1,117.7 h, 720 subjects	No open whole-brain encoder of naturalistic EEG at this scale
Number of inverse problems	One: response → representation	Two: scalp → cortical sources (itself ill-posed) → representation
Measured language-decoding evidence	Continuous language reconstructed from fMRI [4]	Same decoder: 67% character error with EEG vs 32% with MEG [6]
Open naturalistic language data	LeBel 2023: 8 subjects, 85.8 h of stories	No open simultaneous EEG–fMRI set with sustained speech
Temporal resolution	1.5–2 s samples, ≈5 s haemodynamic lag (cost)	Milliseconds (EEG advantage)
Cost and portability	Scanner-bound (cost)	Wearable, low cost (EEG advantage)

Decision. fMRI is where the question can be answered today: it has the spatial detail, the data and the only deep encoder large enough to invert. EEG's real advantages (speed, cost) matter for a deployed device, not for measuring how invertible an encoder is. The method transfers to EEG once a comparable EEG encoder exists.

Four explanations for incomplete recovery

Hypothesis	Predicted signature	Test	Verdict
H1 Information loss (null space)	Error stays flat on the line between fit-matched solutions	Straight-path error profile, forward passes only	**Rejected**
H2 Non-convexity	Distinct minima at equal fit, separated by a barrier	Init control at matched fit (× 4); path rise; near-truth control	**Supported**
H3 Optimiser artefact (under-convergence)	Recovery keeps rising with budget	Convergence curve; fit-matched inclusion	**Controlled**
H4 Unidentifiable gain	Rescaled x produces the same output	Forward-only rescaling test	**Rejected**
H5 Errors along weak directions	Difference directions have low Jacobian response	Directional sensitivity vs 200 random	**Supported**

Validation built into the method

Risk to validity	Safeguard	Outcome
Measuring the optimiser, not the operator	Converge first; independent runs to schedule end	Plateau ≈ 0.94 by 32,000 steps
Under-converged runs posing as alternative solutions	Only blind solutions within 2× of the best blind fit	Gaussian s.d. 5 excluded (3.2× worse fit)
One trajectory being special	Replicate across 4 independent ground truths	min pairwise 0.8165 ± 0.0068
Inference instead of demonstration	Straight-path error between fit-matched solutions, 4 truths	Distinct pairs rise 2.4–8.2× (median 4.8×): H1 rejected
Solutions that were told the answer	Near-truth starts used only as a control	Deeper minimum, 4.2 ± 0.3× lower error
ROI size confound	Random vertex sets of identical size	Paired comparison per network
Random-seed luck in controls	Second random seed for two networks	Δ between seeds ≤ 0.013
Platform dependence	Whole-cortex anchor re-run on CPU vs earlier GPU	0.836 vs 0.836 (GPU)

Constraints, alternatives and trade-offs

Constraint	Effect	Trade-off adopted
No local GPU	Large models cannot run on laptops	Colab Pro T4 GPU for GPU work; CPU for regional analysis; Llama in half precision
12 GB RAM / 15 GB GPU	Llama-3.2-3B and w2v-BERT cannot load together	One story per process; audio and text in separate passes
Released pipeline not differentiable	Raw stimuli cannot be optimised	Invert the aggregated representation x
No ground truth in real fMRI	Recovery cannot be measured directly	Exact synthetic evaluation + identification at a known chance level
Only an averaged subject layer	No per-subject analysis	Invert the population model; stated as a limitation
Synthetic x in a random subspace	Absolute values may not transfer	Real-data validation confirms the qualitative pattern

Alternatives from Report I. (1) End-to-end supervised decoder: needs large per-subject data and cannot explain failures. (2) Inversion of a pretrained encoder: selected. (3) Simultaneous EEG–fMRI fusion decoder: blocked, since no open simultaneous dataset contains sustained language.

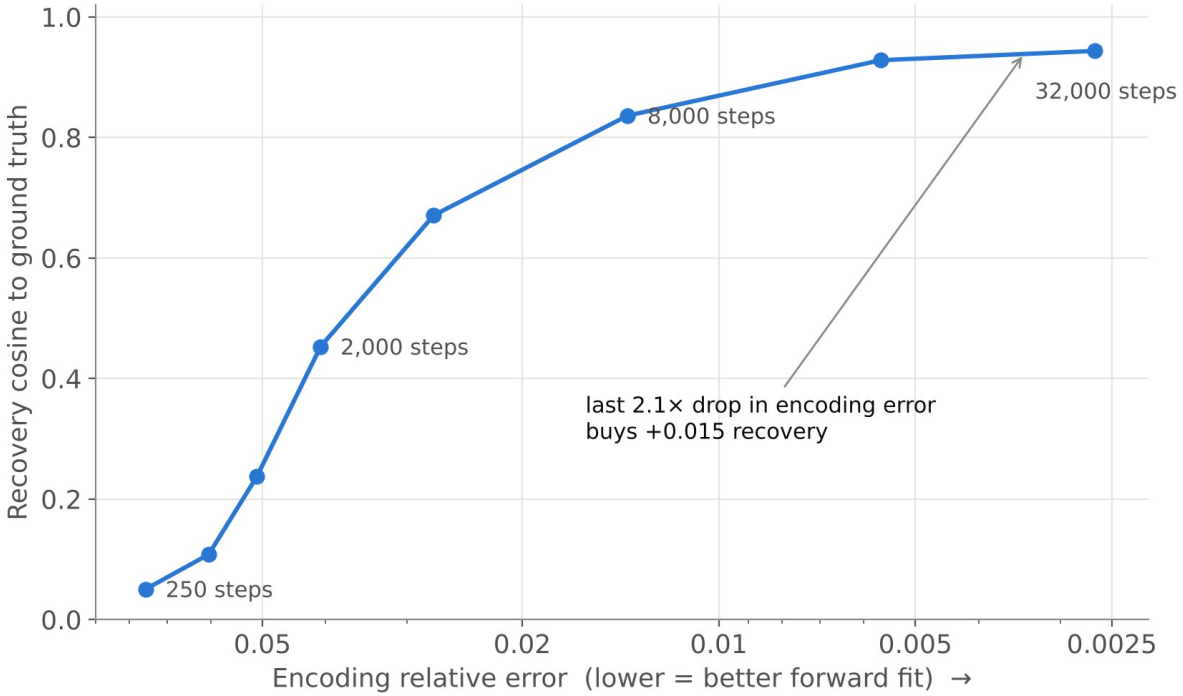
Hardware and software tools

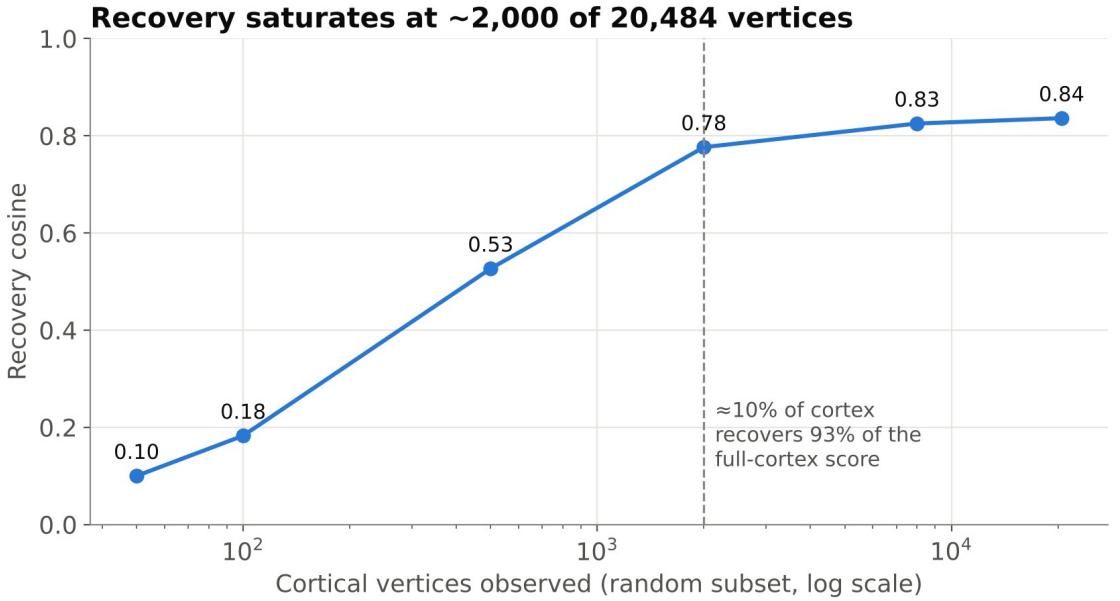
Category	Tool	Use
Compute (GPU)	Google Colab, NVIDIA T4 (15 GB), 12 GB RAM	Synthetic characterisation, real-data features and inversion, verification
Compute (CPU)	Linux cloud CPU	Regional analysis, surface projection, recorded multi-start run, figures
Deep learning	PyTorch, HuggingFace Transformers and Hub	Autograd through the frozen encoder; Llama-3.2-3B, w2v-BERT 2.0
Encoder	TRIBE v2 code and weights (Meta FAIR, CC BY-NC 4.0)	Frozen forward operator
Neuroimaging	nilearn, nibabel, ANTsPy, pycortex transforms	Atlas, surfaces, registration, surface projection
Scientific Python	NumPy, SciPy, scikit-learn, h5py, Matplotlib	Analysis, ridge baseline, figures
Data	OpenNeuro ds003020 (LeBel et al., 2023)	Real naturalistic fMRI
Delivery	GitHub, Colab notebooks, WebGL demo page, PowerPoint, Word	Release, live demo, reviews and report

Results and interpretation

Why recovery is incomplete, and where in cortex meaning is recoverable

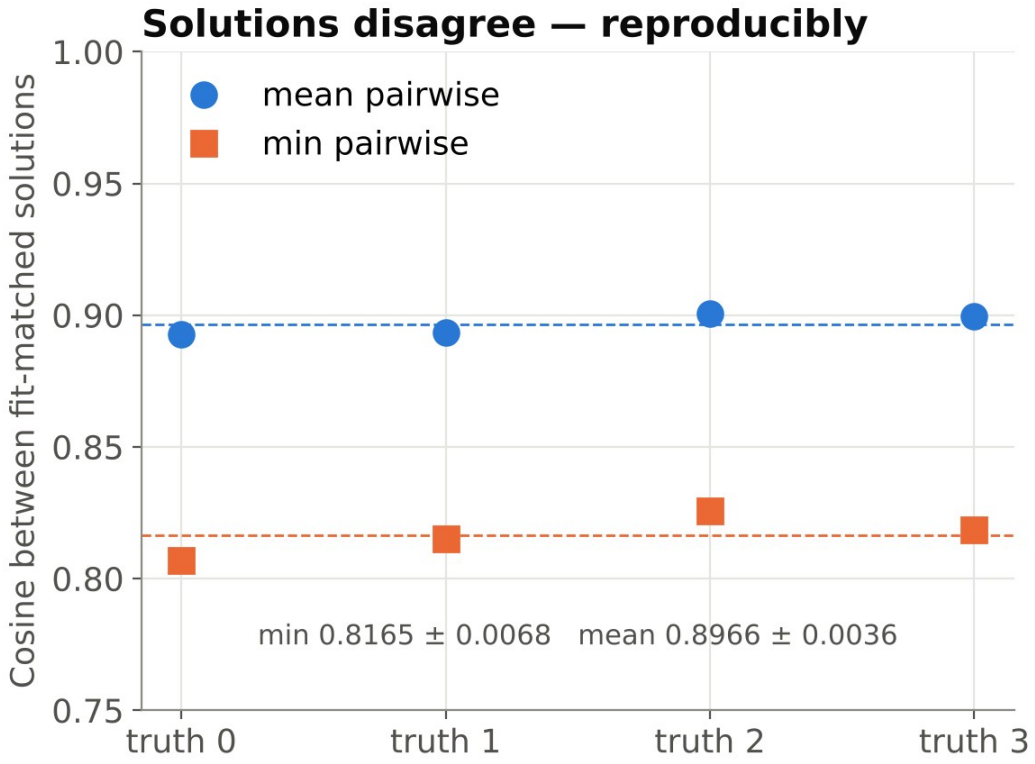
Recap: encoding and decoding dissociate

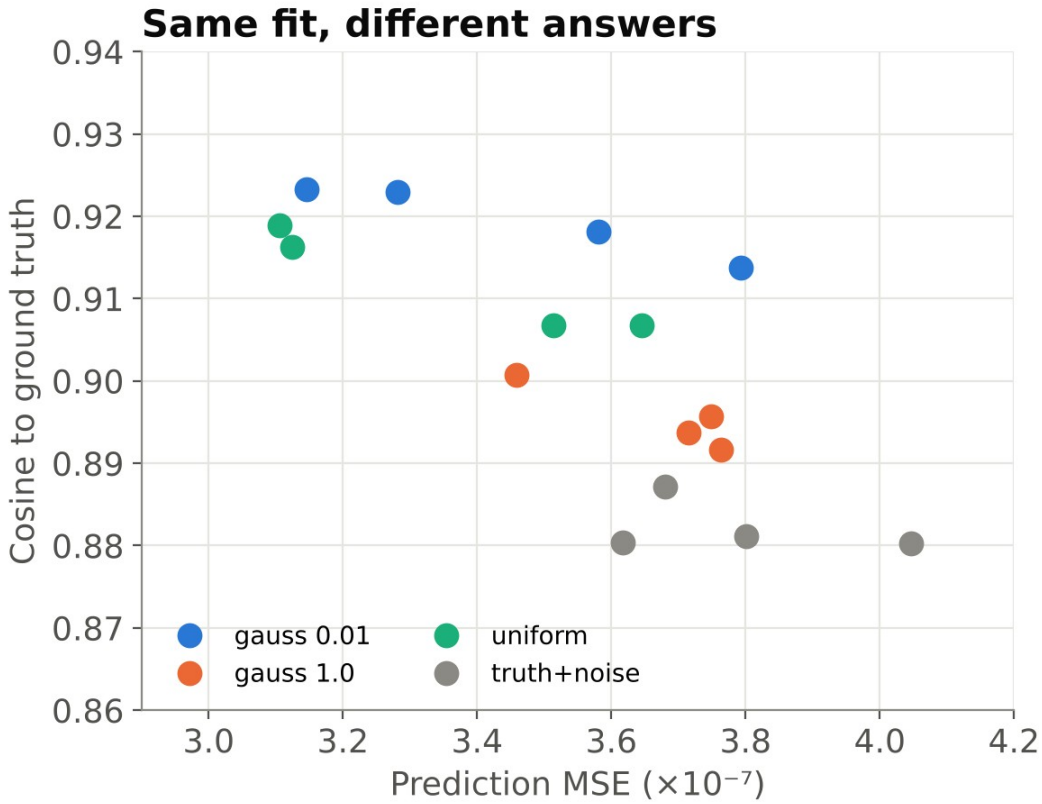




Encoding error falls 28.4× while recovery saturates at 0.944. Recovery saturates near 2,000 random vertices. Both established at Review II.

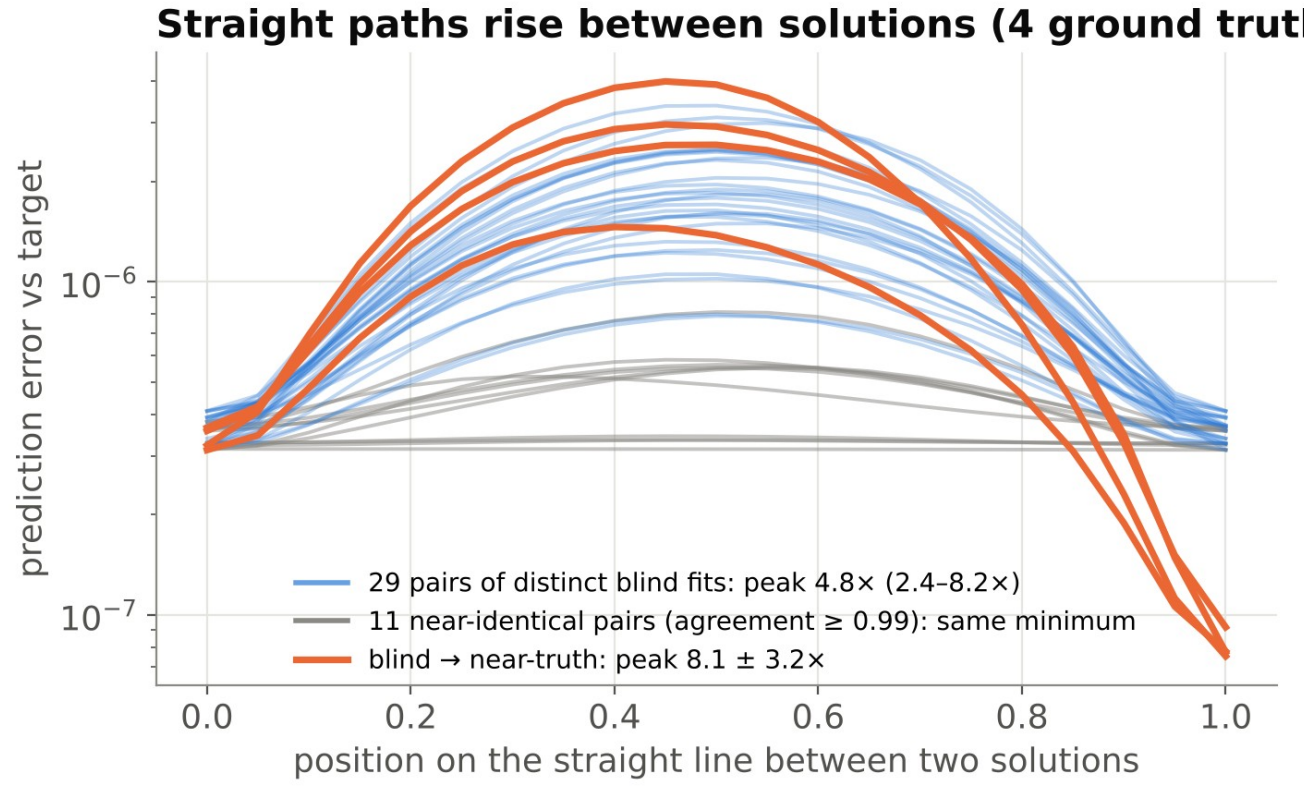
Same fit, different answers: the inverse is non-convex

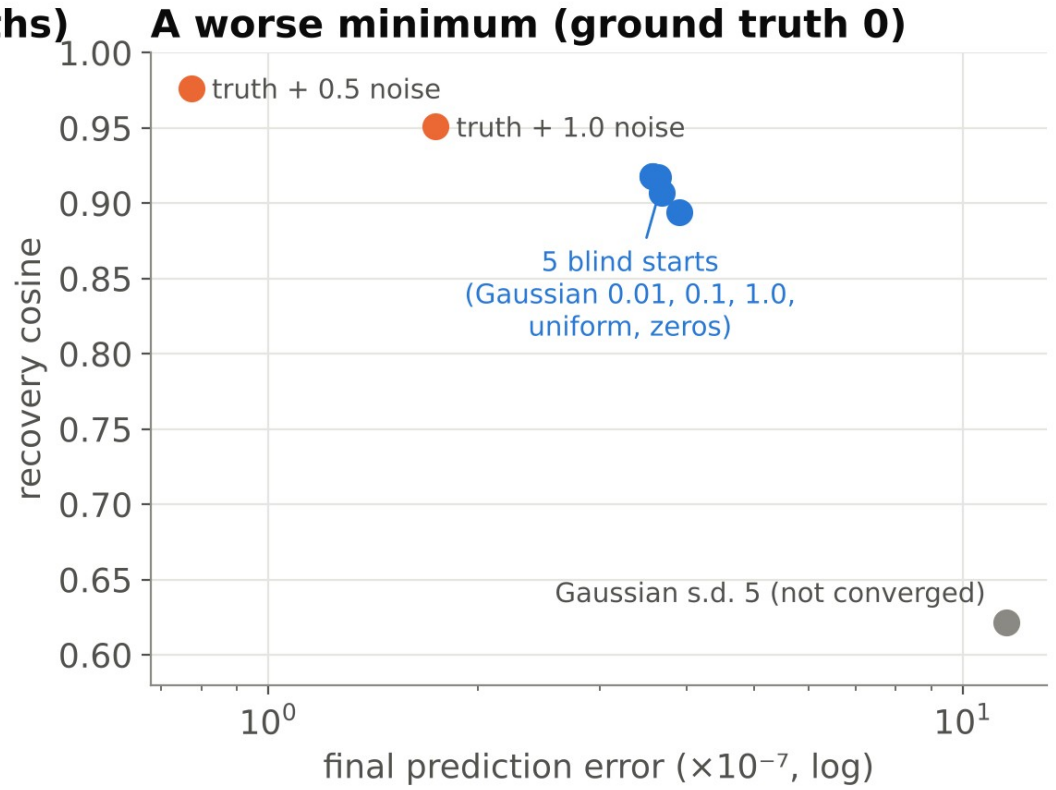




Ground truth	Best fit	Mean pairwise	Min pairwise	Worst pair
0	3.51e-7	0.8928	0.8068	uniform vs truth+noise
1	3.65e-7	0.8934	0.8151	uniform vs truth+noise
2	3.11e-7	0.9007	0.8257	uniform vs truth+noise
3	3.13e-7	0.8996	0.8186	uniform vs truth+noise

Barriers, not a null space (verified on the real model)





Result. Four ground truths, 40 pairs: between every pair of distinct blind fits the error rises 2.4–8.2× (median 4.8×); near-identical pairs share one minimum. A null space would stay level. Near-truth starts find a minimum with 4.2× lower error, 8.1 ± 3.2× barrier away.

Correction. Our earlier perturbation test (2.6× → 0.6×) paired an unconverged solution (Gaussian s.d. 5). It is withdrawn; the selection rule is fixed and the test was re-run on a T4 on 19–20 Oct 2026 for four ground truths.

Blind starts settle in a worse minimum (ground truth 0)

Start	Told the truth?	Final error ($\times 10^{-7}$)	× best blind	Recovery cosine
Gaussian, s.d. 0.01	no	3.63	1.02	0.917
Gaussian, s.d. 0.1	no	3.57	1.00	0.918
zeros	no	3.57	1.00	0.918
uniform on [−1, 1]	no	3.68	1.03	0.907
Gaussian, s.d. 1.0	no	3.90	1.09	0.894
Gaussian, s.d. 5.0	no	11.5 (not converged)	3.23	0.621
truth + 1.0 noise	yes (control)	1.74	0.49	0.951
truth + 0.5 noise	yes (control)	0.77	0.22	0.976

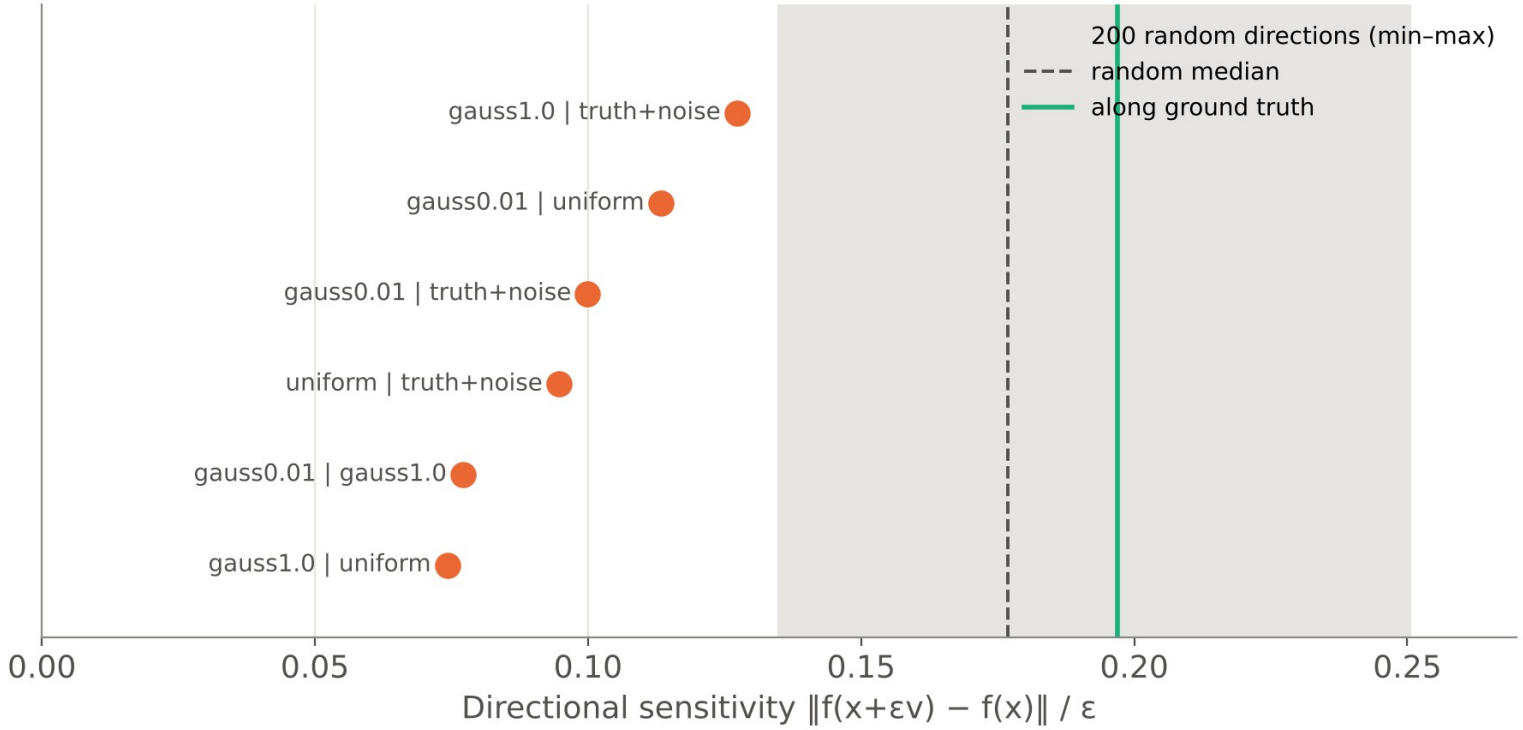
What it means

Every start a real decoder could use (no knowledge of the answer) ends at almost the same error, $3.6–3.9 \times 10^{-7}$, and agrees with the others only at 0.89–1.00.

A deeper minimum exists: starting near the truth reaches 4.6× lower error and recovery 0.976 (4.2 ± 0.3× over four truths).

Between them the error rises 7.2× here (8.1 ± 3.2× over four truths), so gradient descent cannot slide from one to the other. Better search, not a better encoder, is what decoding needs.

Errors concentrate in the operator's weakest directions



2.15×

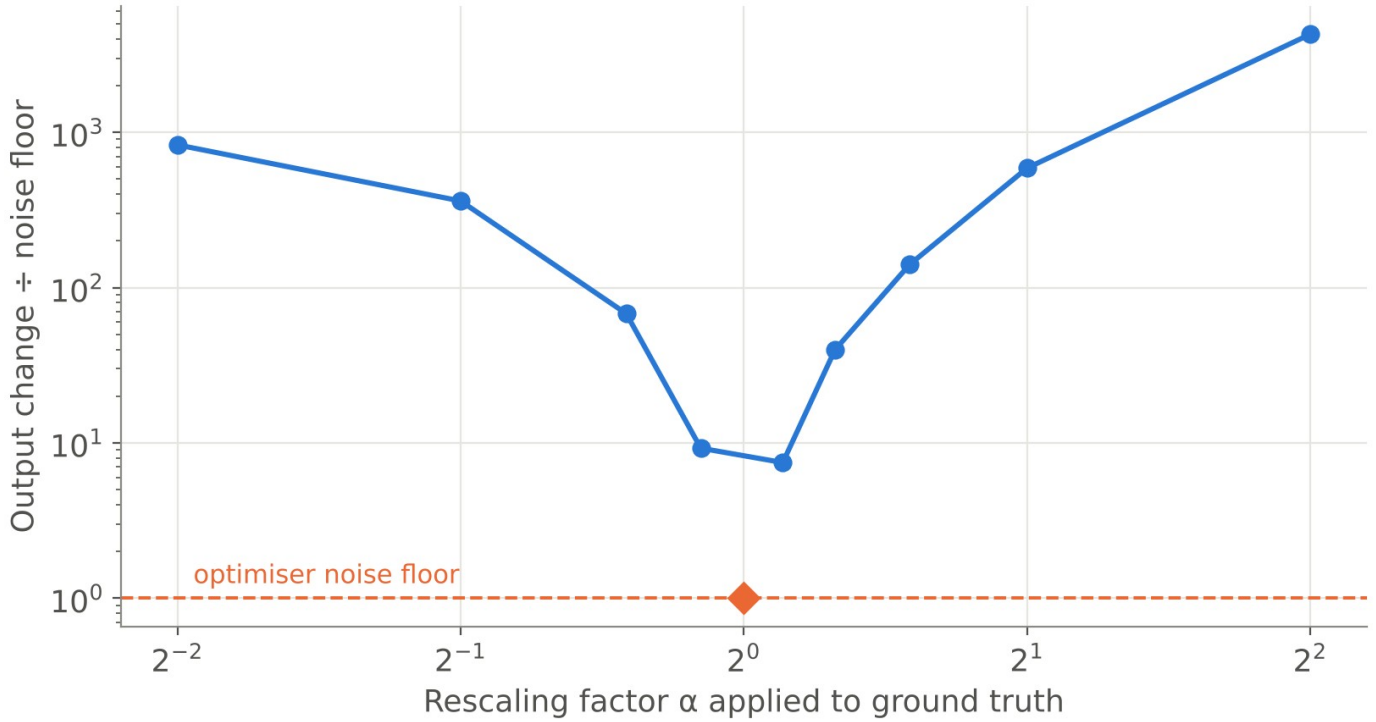
lower sensitivity along d than the random median, mean of 4 truths (± 0.17; 1.9× originally, shown)

1.68×

along the actual error $\hat{x} - x$ (± 0.17, 4 truths); d below all 200 random directions

Anisotropic ill-conditioning: what the inverse fails to pin down is what the operator responds to least (corrected pair s = 0.081 vs random 0.122–0.239; actual error 1.8× below).

Gain is identifiable; error is not a single scale factor

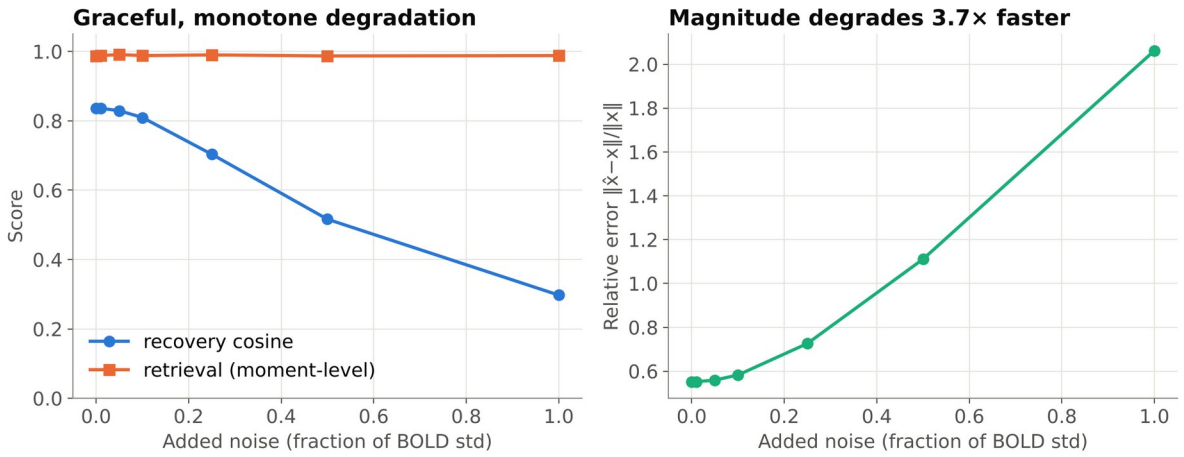


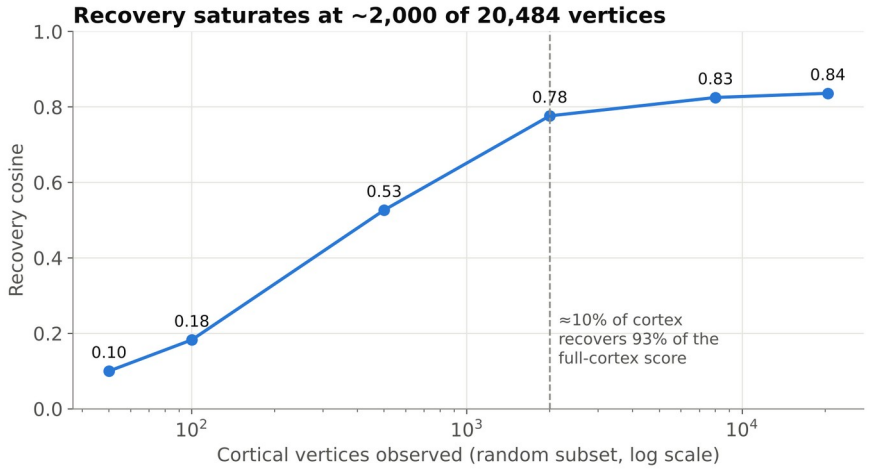
Quantity	Value
Direction cosine	0.8362
Mean norm ratio $\|\hat{x}\|/\|x^*\|$	0.747 (sd 0.046)
Norm correlation over time	0.915
Relative error, raw	0.5516
Relative error after optimal rescale	0.5675

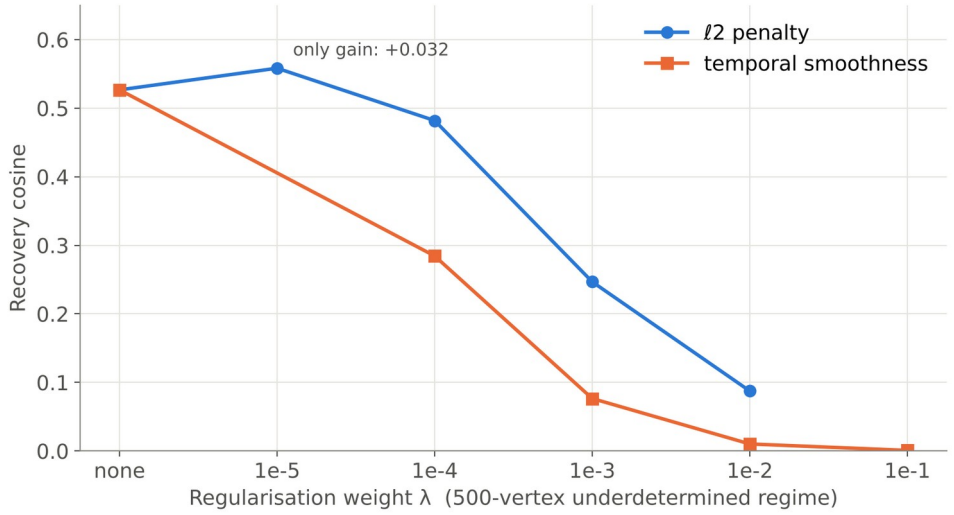
Rescaling makes error slightly worse, so the shortfall is not a global gain. Even ±10% scaling is 7–9× above the noise floor.

Two simple explanations excluded: the decoder is not merely losing the overall magnitude of the representation.

Robustness: noise, observed cortex and priors

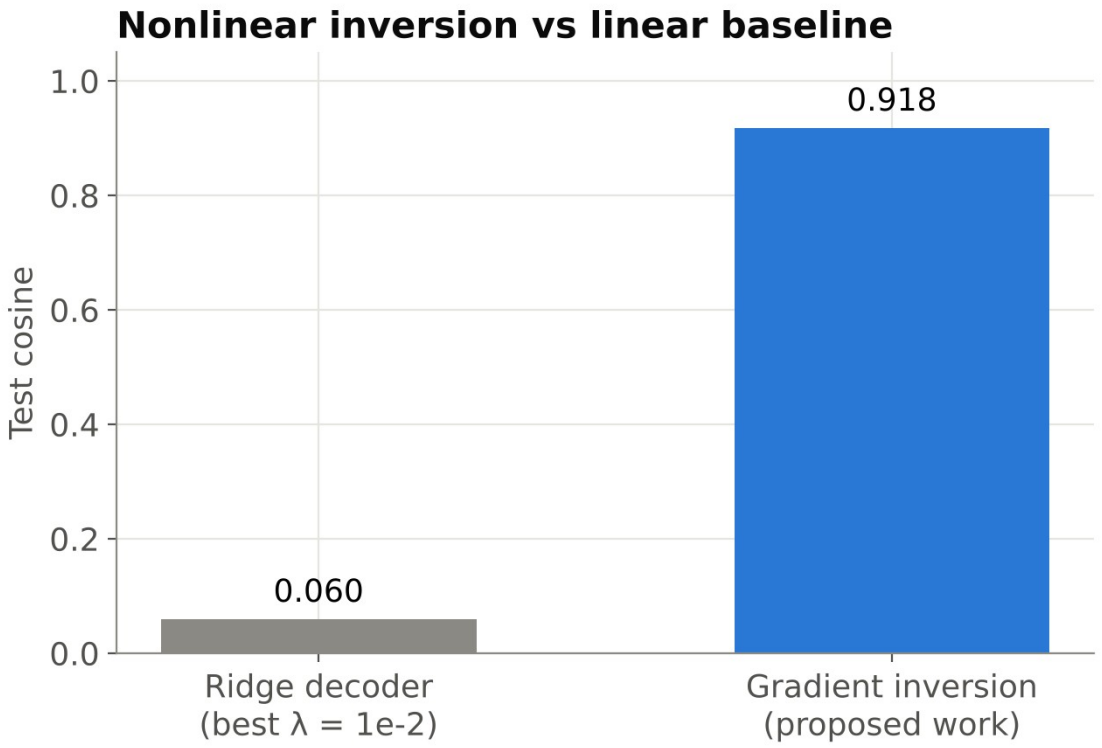






Test	Result
Noise 0 → 100% of BOLD s.d.	recovery 0.836 → 0.298, monotone; retrieval ≈ 0.99
Observed cortex 50 → 20,484	0.100 → 0.836; 2,000 random vertices reach 0.777
ℓ_2 prior	best +0.032 at $\lambda = 10^{-5}$, harmful above
Smoothness prior	0.527 → 0.0008
Stimulus manifold	white 0.947 vs smooth 0.942
Rate of change, 64×	0.836–0.858, no trend

Nonlinear inversion vs a linear decoder



Ridge λ	Test cosine
1e-2	0.0599
1e-1	0.0310
1e0	0.0205
1e1	0.0133
1e2	0.0103

Caveat stated plainly. Ridge (240 train / 60 test pairs) predicts the time-averaged representation from time-averaged BOLD, a strictly easier target than the time-resolved trajectory inversion recovers. Read 0.060 as a floor; the 15× gap exceeds what the mismatch can explain.

What did not work, reported as such

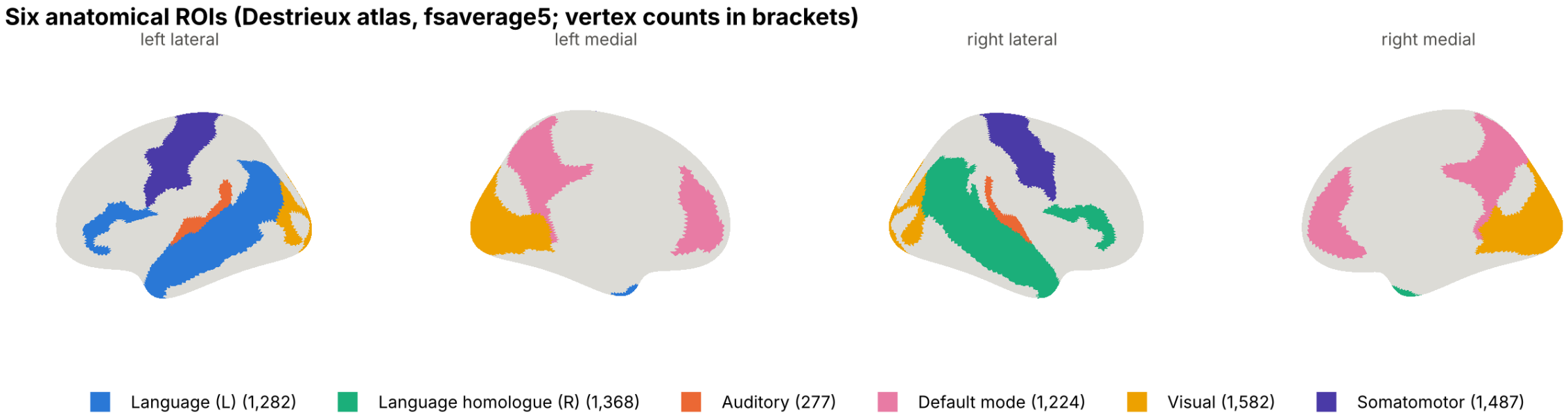
Experiment	Outcome	Why	How we report it
Optimiser control (SGD, L-BFGS)	Fits 1.4e-4 / 2.8e-5 vs Adam 3.5e-7	Hyperparameters did not converge	Invalid control; limitation
Bandwidth sweep	Flat 0.838–0.858 over 64×	No HRF operator in the release	Structural null result
Stimulus manifold	0.947 vs 0.942	Operator indifferent to prior	Null result
Smoothness prior	0.527 → 0.0008	Wrong prior for semantics	Negative result
Original barrier test	Paired an unconverged solution	Pair chosen among all 8 starts	Withdrawn; re-run correctly

Why this slide exists: the non-convexity claim does not rest on the failed optimiser control. It rests on the straight-path barrier and sensitivity tests (forward passes only), the near-truth control and replication across four ground truths. A converged second optimiser would strengthen it, and is listed as future work.

Regional analysis (neurobiology track)

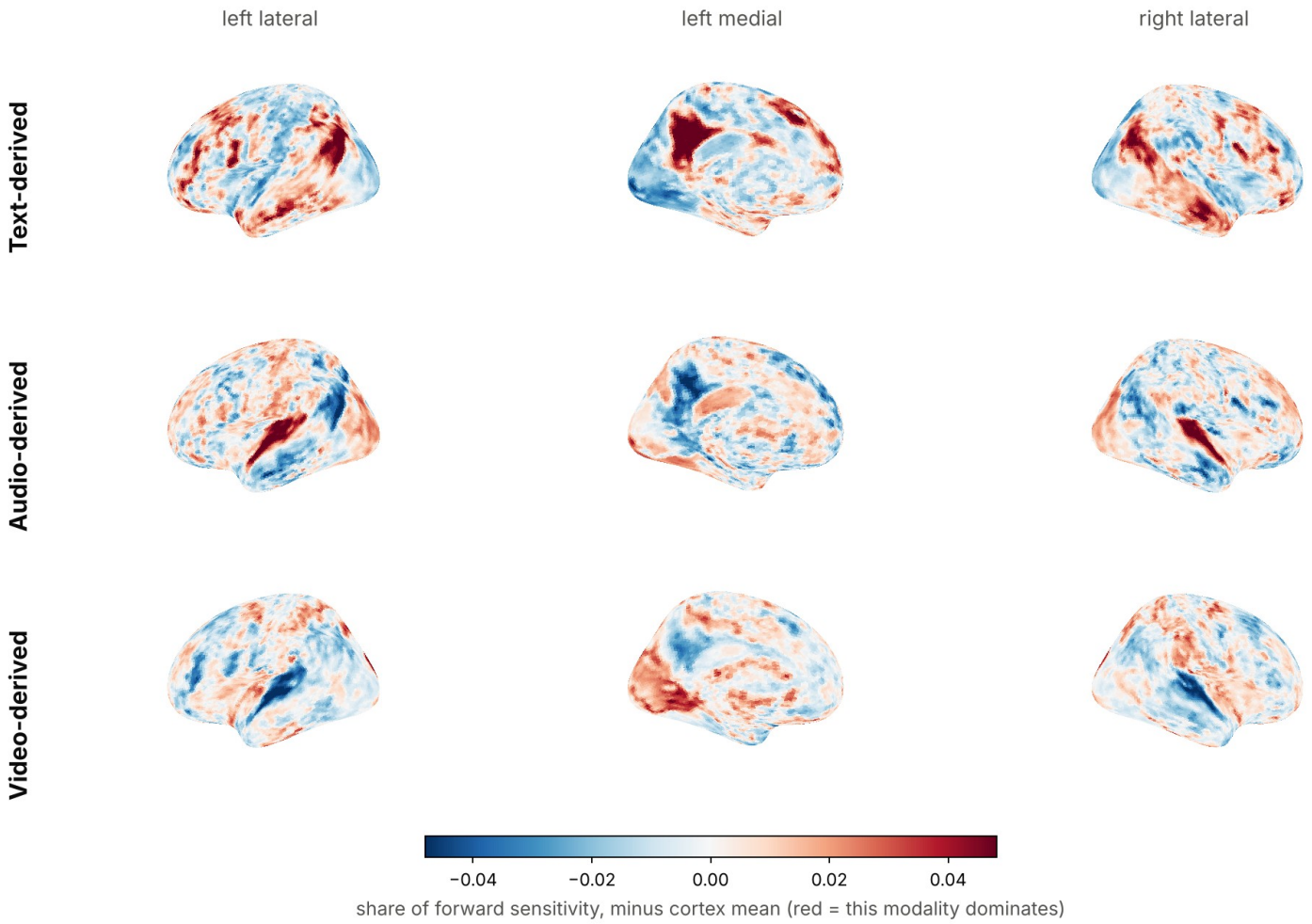
Which networks the operator listens to, and where meaning is recoverable

Six anatomical networks on fsaverage5



Network	Destrieux labels	Vertices
Language (L)	IFG opercular/triangular/orbital, STG lateral, STS, MTG, temporal pole, angular gyrus	1,282
Language homologue (R)	Same labels, right hemisphere (lateralisation test)	1,368
Auditory	Heschl's gyrus, transverse temporal sulcus, planum temporale	277
Default mode	Precuneus, posterior cingulate (dorsal, ventral), anterior cingulate, subparietal	1,224
Visual	Occipital pole, cuneus, calcarine, middle/superior occipital, lingual	1,582
Somatomotor	Precentral, postcentral gyri, central sulcus (low-semantic control)	1,487

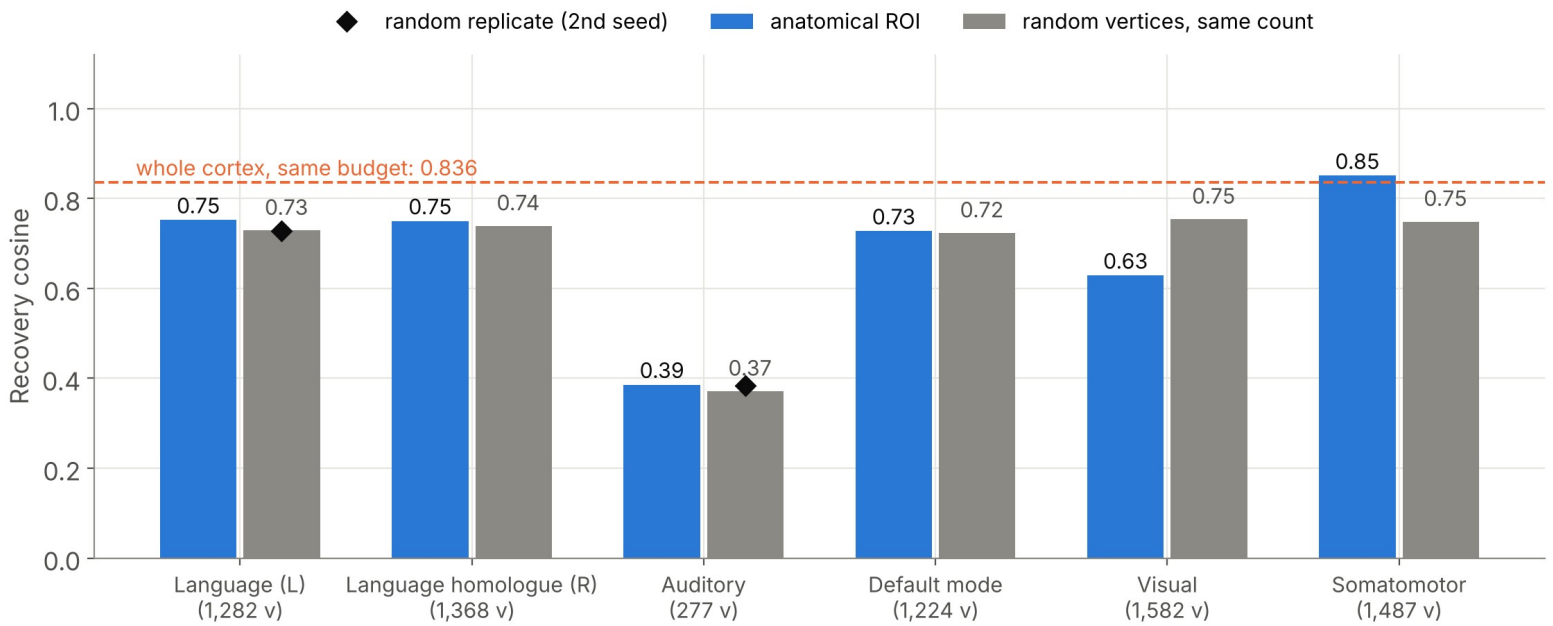
The operator's functional topography matches anatomy



Network	text z	audio z	video z
Language (L)	**+26.6**	-12.5	-18.5
Language homologue (R)	**+28.5**	-10.6	-23.3
Auditory	-3.8	**+34.7**	-32.8
Default mode	**+26.9**	-15.1	-16.1
Visual	-38.4	**+7.5**	**+40.7**
Somatomotor	-3.5	**+6.9**	-3.5

Each network is dominated by the modality it should be: language and default-mode by text-derived dimensions, auditory cortex by audio, visual cortex by video. Somatomotor is least sensitive overall (0.73× cortex mean).

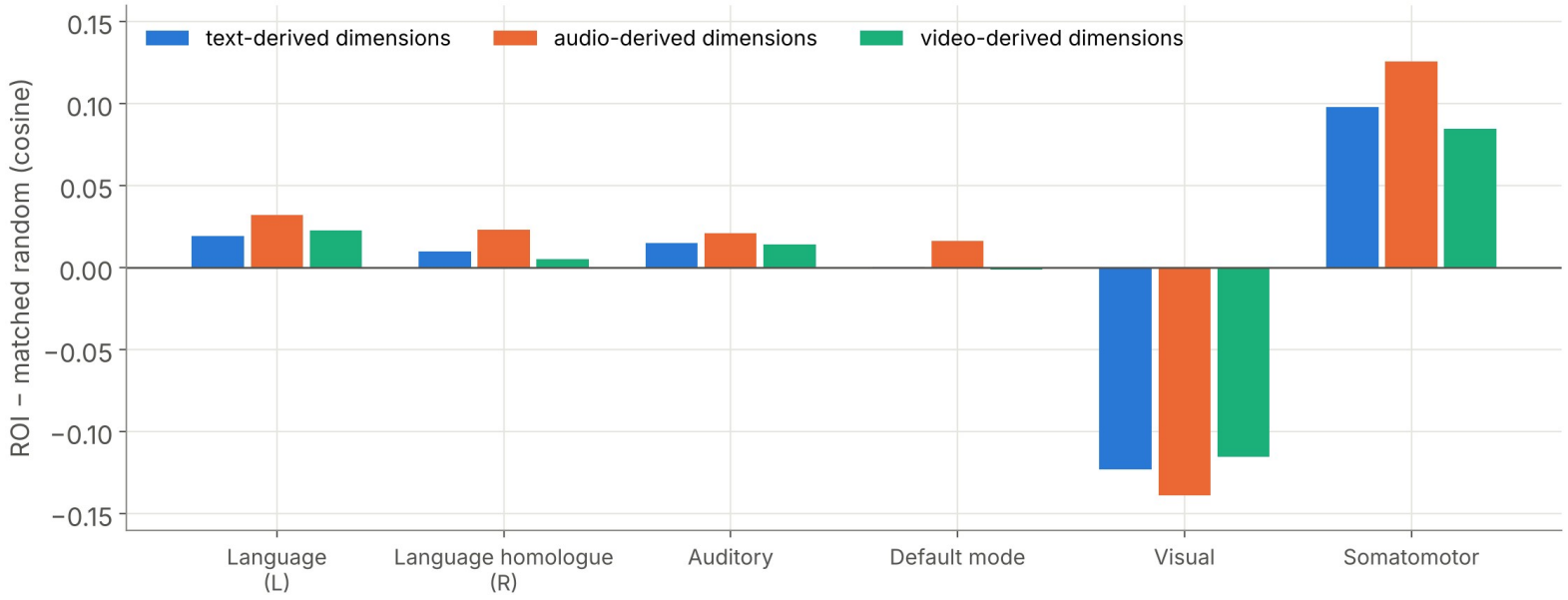
Where you look matters: visual cortex hurts, somatomotor helps



Network	ROI	Random	Δ
Language (L)	0.752	0.729	+0.023
Language homologue (R)	0.750	0.739	+0.011
Auditory	0.386	0.371	+0.015
Default mode	0.727	0.723	+0.004
Visual	0.629	0.754	−0.125
Somatomotor	0.851	0.748	+0.102

Visual cortex recovers 0.125 worse than random vertices of the same count; somatomotor 0.102 better, beating the whole cortex at the same budget (0.851 vs 0.836). Both effects are ~10× the control seed spread (≤ 0.013). Left language +0.023; homologue, auditory, default mode within noise.

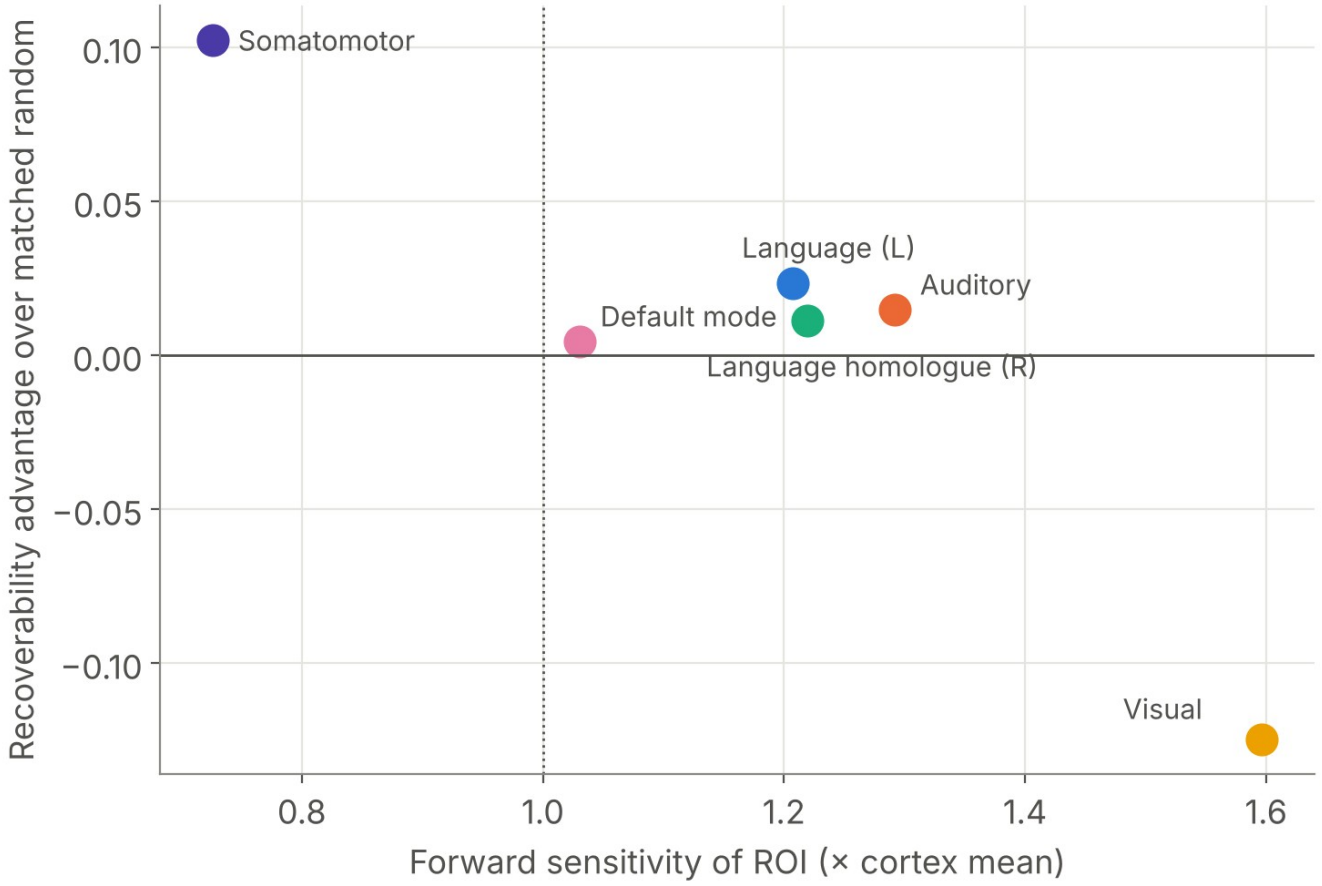
The regional effect is not modality-specific



Split recovery into the text-, audio- and video-derived thirds of x. If a network helped only with the modality it encodes, visual cortex would recover video dimensions well. Instead, every network shifts all three thirds together: visual cortex loses text, audio and video alike; somatomotor gains all three.

Regional recoverability is a property of how a region's response is shaped, not of which modality it is tuned to.

Predictable is not recoverable

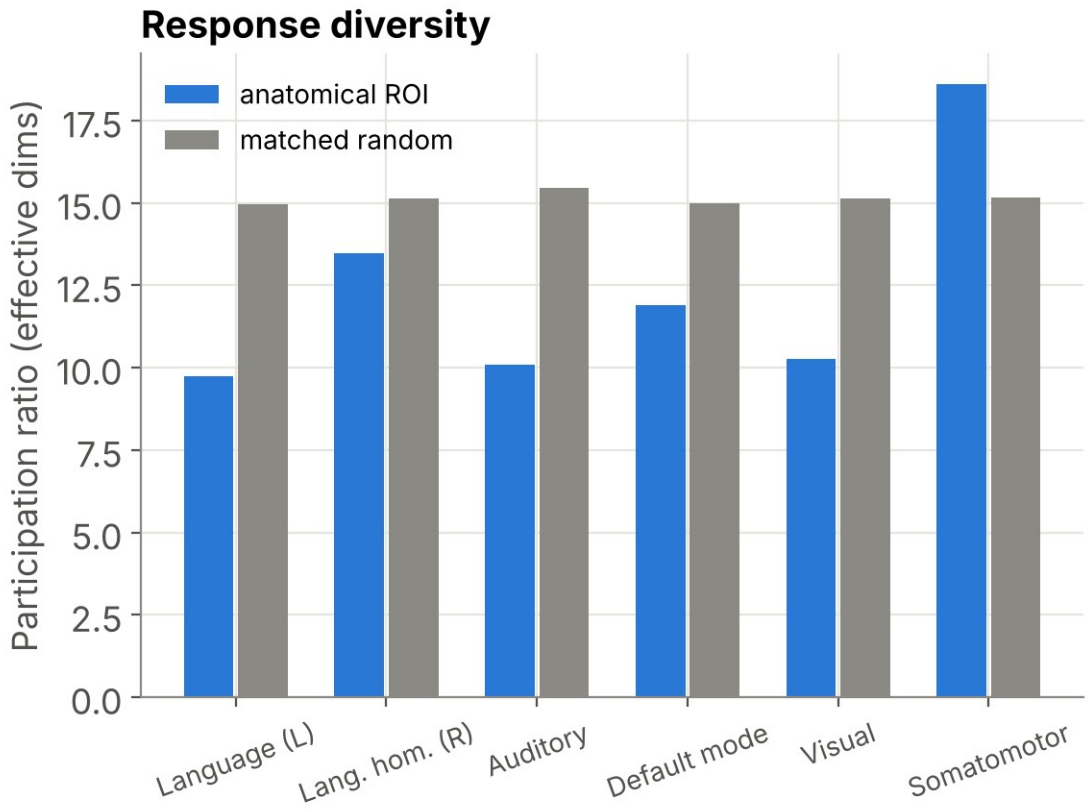


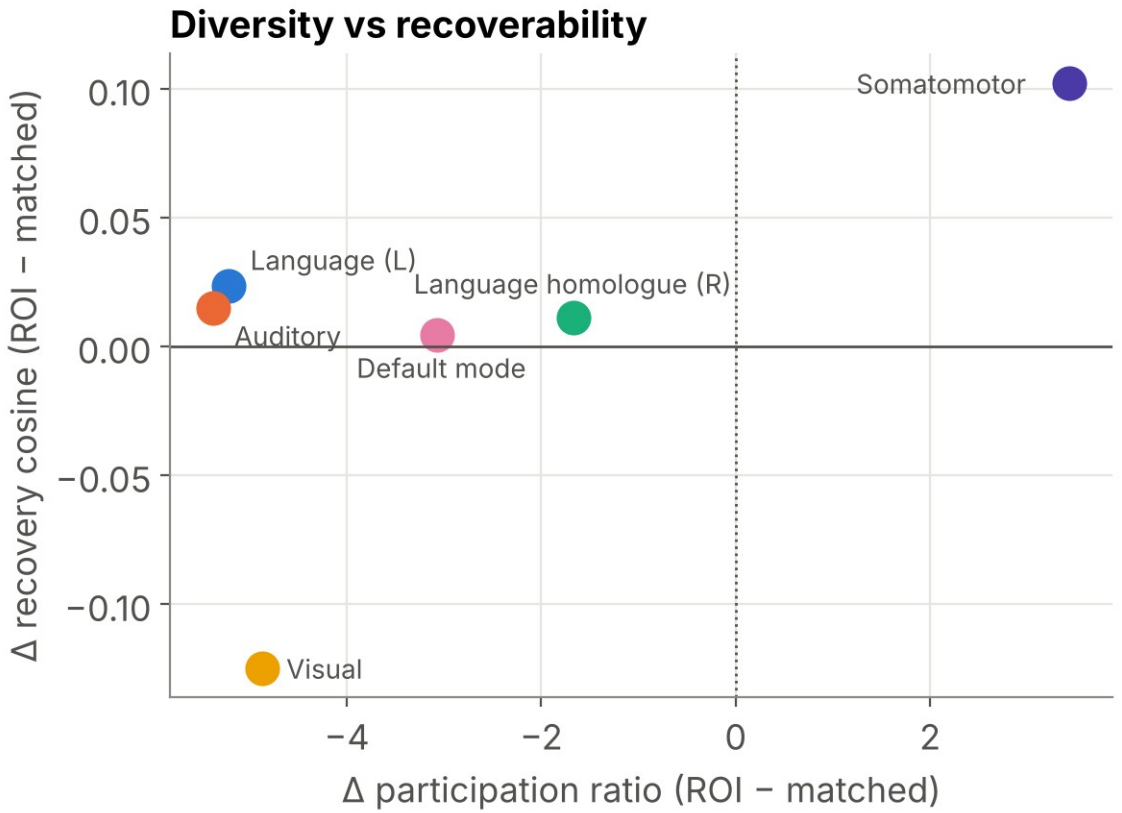
Visual cortex responds most strongly to the representation (1.60× cortex mean) and is the worst place to decode it from. Somatomotor responds least (0.73×) and is the best.

Partial mechanism: response diversity. Somatomotor's response spans the most independent directions (participation ratio 18.6 vs 15.2 matched; top direction carries 15% of variance). Visual responses are strong but concentrated (PR 10.3; top direction 27%).

Language and auditory are also concentrated yet near their controls, so diversity explains the extremes, not every network. Six networks are too few for a correlation claim.

A partial mechanism: response diversity





Somatomotor spans the most independent input directions (PR 18.6 vs 15.2) and recovers best; visual is concentrated (PR 10.3) and recovers worst. Language and auditory are concentrated too but near their controls: diversity explains the extremes only.

What limits semantic decoding from fMRI

Not destroyed information	No null space: the error rises 4.8× between distinct equal fits, 8.1× towards a deeper minimum (4 truths).
Non-convexity	Equally good fits disagree at 0.82 cosine, reproducibly; the answer depends on where the search starts.
Anisotropic conditioning	Disagreement lives in directions the operator barely responds to (0th percentile).
Measurement noise	Graceful loss of direction, steep loss of magnitude (rel. error 0.55 → 2.06).
Where you look	Strong responses are not the best evidence: visual cortex decodes worst, somatomotor best (−0.125 / +0.102 vs matched).

Tabulation of all results

Claim	Evidence	Key number	Status
Encoding ≠ decoding	8 independent budgets	28.4× vs +0.015	Established
Inverse is non-convex	Fit-matched init control, 4 truths	min pairwise 0.8165 ± 0.0068	Established
No null space	Straight-path error, 40 pairs, 4 truths	4.8× (2.4–8.2×); 8.1 ± 3.2× to deeper	Verified ×4
Errors in weak directions	200 random directions, 4 truths	2.15 ± 0.17× below median	Verified ×4
Gain identifiable	Forward rescaling	±10% → 7–9× floor	Established
Stable under noise	7 noise levels	0.836 → 0.298 monotone	Established
≈10% cortex suffices	6 extents	0.777 at 2,000 vertices	Established
Beats linear decoding	Ridge baseline	0.918 vs 0.060	Indicative
Anatomical topography	Sensitivity maps, 6 networks	z +27 to +41 in matching modality	Established
Regional recoverability	ROI vs matched random	Visual −0.125, somatomotor +0.102 vs matched	Established (1 truth)
Real-data decodability	LeBel 2023, held-out story	4/5 segments (chance 2.2%), cosine 0.02	Established
Multi-start on the real model	Recorded CPU run, 3 starts	R^2 0.995, recovery ≤ 0.51, agreement 0.37	Demonstrated

Validation on real naturalistic fMRI

LeBel et al. 2023 · subject UTS03 · held-out story wheretheressmoke

Real-fMRI pipeline

Stage	What was done	Check
Data	UTS03, LeBel et al. 2023 (OpenNeuro ds003020): 6 stories, 66.6 min, 10,363 words, TR 2 s; held-out story = mean of 10 repeats	Held-out story is TRIBE v2's own test split
Registration	MNI → subject T1 by ANTs SyN; T1 → functional voxels by the pycortex transform	Visual QC of overlays
Surface projection	TRIBE v2 projector settings: 3-mm ball, 20 points per vertex, trilinear in the cortical mask → sparse 20,484 × 95,556 sampler	Homotopic r 0.635 vs −0.017 heterotopic (> 97% of pairings); scrambled control at chance
Features	w2v-BERT 2.0 audio + Llama-3.2-3B text, TRIBE v2's own pipeline with the dataset's word timings; 45 segments of 100 s	Forward re-entry r = 1.0000
Alignment	Lag maximising prediction–response r on the 5 training stories	Peak at 4 s
Evaluation	Encoding r vs split-half ceiling; forward identification of 20-s windows (194); inversion of 100-s segments ranked among 45; noise-free oracle	Chance 0.5% and 2.2%

Real fMRI: results on the held-out story (LeBel 2023)

OpenNeuro ds003020
UTS03 · 6 stories · TR 2 s

→

Surface projection
validated sampler → fsaverage5

→

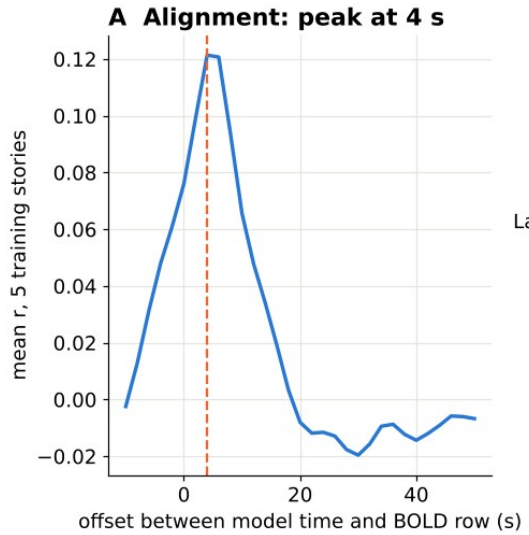
Stimulus features
LLaMA-3.2-3B text + w2v-BERT audio

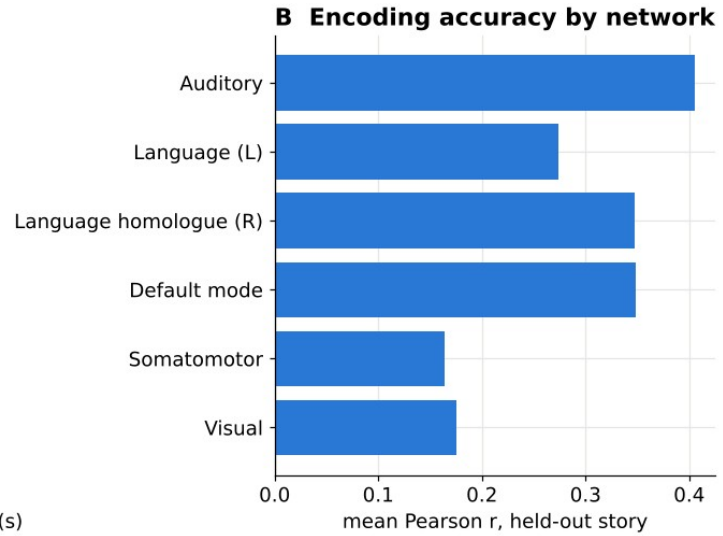
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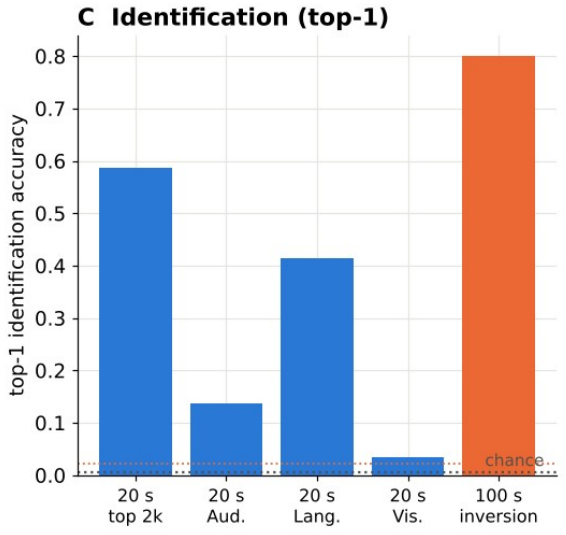
Inversion
held-out BOLD → $\hat{x}$ through frozen f

→

Identification
rank the true moment among all stories







0.48
encoding r, top 2,000 vertices (56% of noise ceiling)

59%
forward identification of 20-s windows, chance 0.5% (194 windows)

4/5
held-out segments identified by inverting real BOLD, chance 2.2%

0.02
cosine of $\hat{x}$ to the true x: the moment is identifiable, the representation is not recovered

Real fMRI: prediction accuracy by story and network

Story	Top 2,000	Aud.	Lang. L	Lang. R	DMN	SomMot	Visual	Whole
wheretheressmoke (held out)	**0.48**	0.41	0.27	0.35	0.35	0.16	0.18	0.20
adollshouse	**0.42**	0.33	0.24	0.25	0.32	0.14	0.14	0.16
avatar	**0.21**	0.17	0.11	0.12	0.12	-0.00	0.09	0.05
legacy	**0.31**	0.20	0.23	0.19	0.18	0.09	0.06	0.12
souls	**0.29**	0.13	0.20	0.19	0.14	0.06	0.08	0.10
undertheinfluence	**0.41**	0.28	0.31	0.27	0.24	0.15	0.26	0.18

Held out and in-distribution alike. On the held-out story the encoder reaches r = 0.48 on the 2,000 vertices chosen from the other stories, 56% of the split-half noise ceiling (0.86). Auditory, language and default-mode networks lead; visual and somatomotor trail, as expected for spoken stories.

Adding language features. Text + audio vs audio only on the held-out story: top-2,000 r 0.48 vs 0.27; default mode 0.35 vs 0.10. One story (avatar) shows no clear alignment peak and is kept only as a distractor, which makes identification harder, not easier.

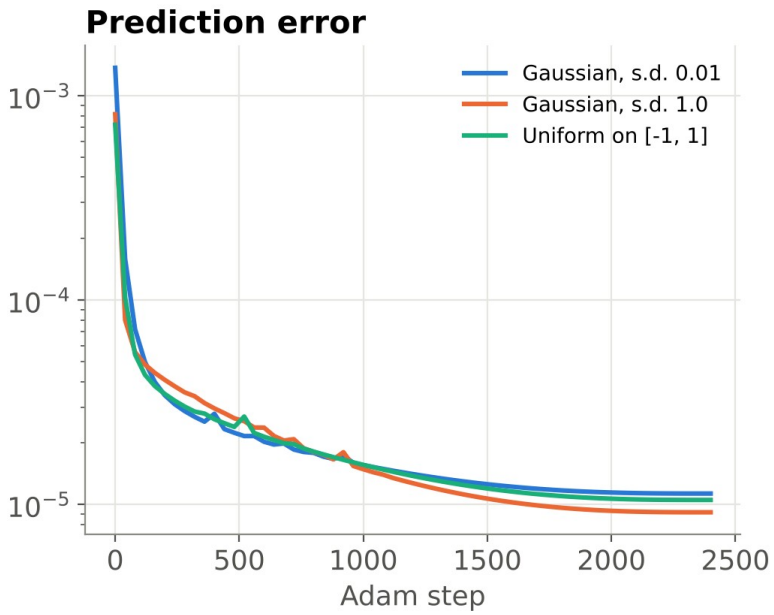
Real fMRI: the inverse identifies, but does not reconstruct

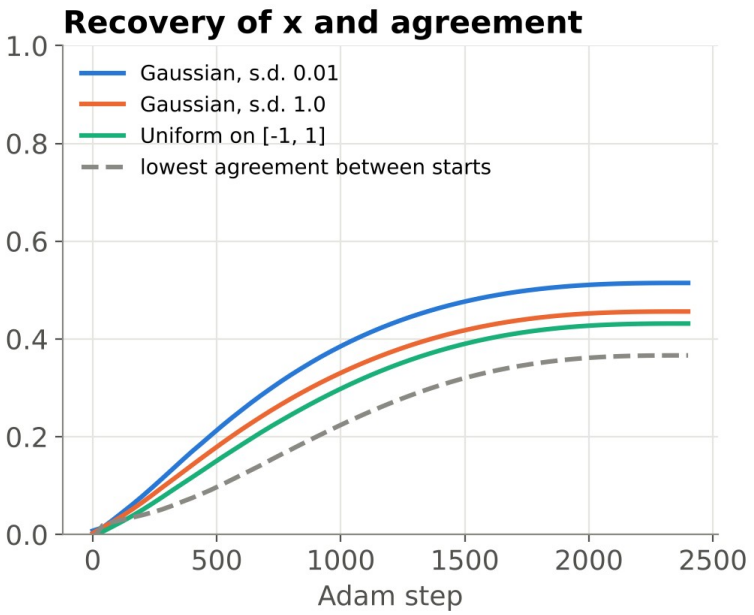
Inversion target	Top-1 identification	Ranks of the true segment	Cosine of $\hat{x}$ to true x	Fit (1 − r)
Measured BOLD, held-out story	4 / 5	0, 0, 6, 0, 0	0.02	0.14
Oracle: model's own prediction	5 / 5	0, 0, 0, 0, 0	0.22	0.0008
Chance	2.2% (1 of 45)	—	—	—

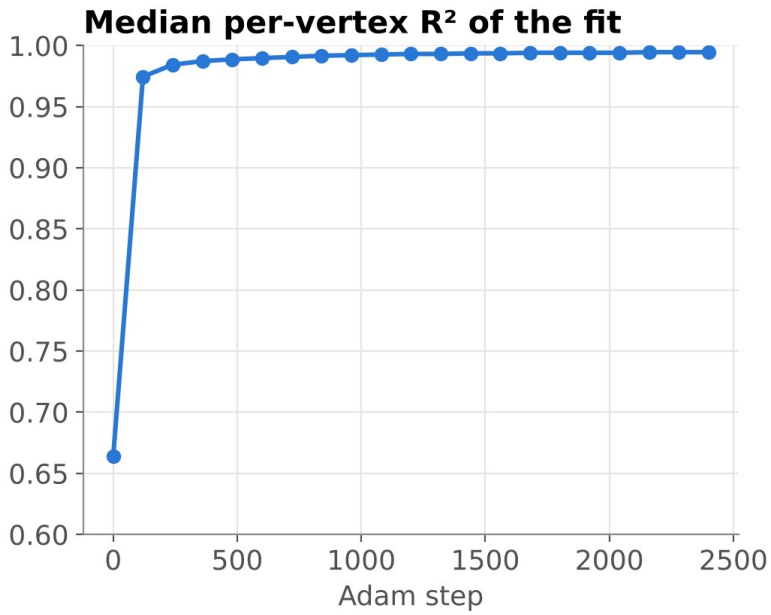
Two limits, separated.
Without noise (oracle), inversion identifies every segment yet recovers only 0.22 of the representation: most of the loss is the operator's conditioning, not measurement noise. With real BOLD, recovery drops further to 0.02 while identification survives (4/5, ranked against all 45 segments of six stories, including the other six of the same story).

This is the synthetic result, reproduced on a real brain: under noise, retrieval stayed near 0.99 while cosine fell. Identification needs only the directions the operator transmits well.

The same experiment on the real model, recorded







0.995

median per-vertex R^2 of the fit across cortex

0.51

best recovery of x after 2,400 steps

0.37

lowest agreement between the three starts

121×

error reduction, start 1

Standards, ethics and SDG justification

Where the work sits against data, reporting and neuro-ethics frameworks

Alignment with relevant standards

Standard / framework	Scope	How this project aligns
BIDS (Gorgolewski et al., 2016)	Neuroimaging data organisation	Real-data phase uses OpenNeuro ds003020, a BIDS dataset; derivatives kept in BIDS layout
IEEE P2794 (draft, active PAR 2022)	Reporting standards for in vivo neural interface research	Report adopts its minimum-reporting approach: model, data, preprocessing, metrics, failures
IEEE P2731 working group	Unified BCI terminology (functional model)	Block diagram uses its functional decomposition: acquisition → features → decoding → output
FAIR principles (Wilkinson et al., 2016)	Findable, accessible, interoperable, reusable data	Seeded scripts, results.json, open model and data; figures regenerated from logs
Licences: CC BY-NC 4.0; Llama 3.2 licence; CC0	TRIBE v2; text encoder; OpenNeuro data	Non-commercial academic use only; gated access obtained
Digital Personal Data Protection Act 2023 (India)	Personal data processing	No personal data processed: synthetic ground truth; de-identified public corpus

Neuro-ethics: what our results imply

OECD Recommendation on Responsible Innovation in Neurotechnology (2019)

Calls for safety assessment and protection of personal brain data. Our limits are a form of safety assessment: they quantify how much a decoder on fMRI can actually recover.

UNESCO Recommendation on the Ethics of Neurotechnology (adopted 12 Nov 2025)

Treats neural data as uniquely sensitive and requires informed consent and protection against misuse. No new neural data was collected in this project.

Implication of our findings

Even with a near-perfect forward model, recovery is incomplete, depends on initialisation, and degrades with noise. Claims of fMRI 'mind reading' should be read against these limits.

SDG justification, target by target

SDG target	Link	Evidence from this project
3.4 promote mental health and well-being	Assistive communication for cognitively intact, motor-impaired patients	Limits tell clinicians what non-invasive fMRI decoding can and cannot deliver; non-convexity points to better search, not bigger encoders
9.5 enhance scientific research and technological capability	Method for characterising neural-interface models	Reusable, open pipeline: differentiable entry into a 177 M-parameter encoder; runs on free compute
10.2 inclusion irrespective of disability	Communication access as a precondition for inclusion	Regional result guides which cortex a future low-cost or partial-coverage system should target

Honest scope: this is enabling research. The SDG contribution is evidence about what is achievable, not a deployed assistive device.

Outcomes, mappings and impact

Deliverables, demonstration, schedule, cost, contributions and programme mapping

Deliverables

Manuscript

IEEE-format paper: “Predictable Is Not Recoverable: Conditioning and Non-Convexity of Inverting a Foundation Brain Encoding Model”. A. Sarkar*, T. Choudhuri* (joint first), S. Nageswaran. Ready for submission.

Final report

48 pages, VIT 2026 Project-I format: 13 chapters, 18 figures, 20 tables, 42 IEEE references, TRIBE v2 data and why-fMRI sections.

A3 poster

VIT template, every box filled; print-ready A3 PDF.

Open repository

github.com/Ivxn404/semantic-decoding-tribe-inversion: all scripts, logged results, figures, demo video, verification results; MIT-licensed code.

Four Colab notebooks

Demo & synthetic suite · real fMRI (text + audio) · 10-minute live demo on TRIBE v2 · verification tests.

Interactive live demo

Ten-scene presenter page: 3-D cortex, in-browser inversion lab, real-run replay, findings, networks, real-fMRI ranking; works offline.

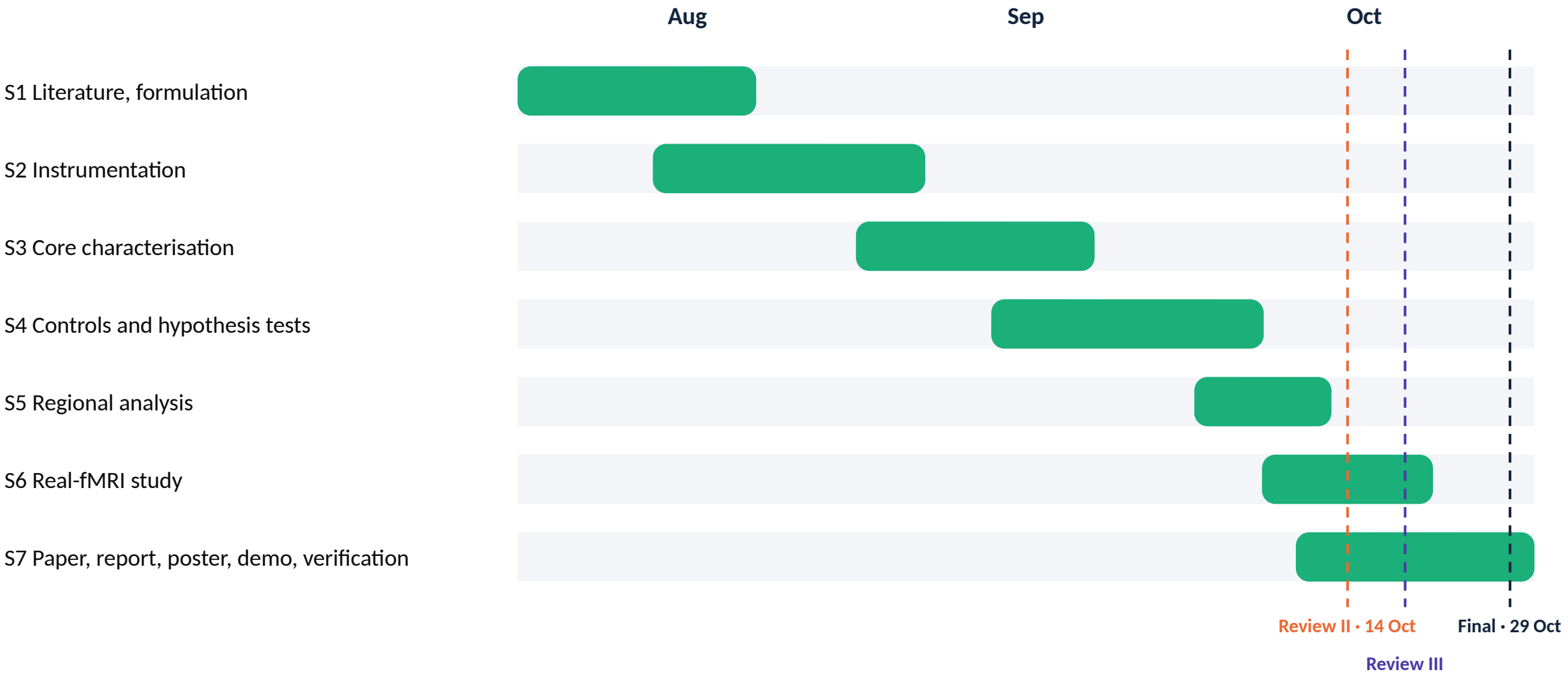
Live demonstration

Scene	What the panel sees	Live?
Question	Rotating cortex coloured by TRIBE v2's measured text / audio / video sensitivity	interactive
Data · Why fMRI	Clickable training and test datasets; fMRI vs simulated EEG at 32 / 64 / 128 electrodes	interactive
Live lab	A small encoder inverted in the browser: starts, spread, noise, observed cortex, prior; sensitivity probe	computed live
Real run	Replay of three starts through frozen TRIBE v2: target, prediction and R^2 maps on the cortex	recorded, real
Findings · Regions	Five findings incl. verified barriers; clickable networks with recovery vs random	interactive
Real fMRI	Lag, encoding and identification charts; held-out segments ranked among 45	animated, real
Colab Live_Demo	The real 177 M model inverted on a T4 in ≈ 3 min, with ridge test and sensitivity probe	computed live

github.com/lvxn404/semantic-decoding-tribe-inversion

Repository, demo page and notebooks

Schedule and milestones: complete



Cost analysis

Item	Estimate in Report I (₹)	Project cost, Aug–Oct 2026 (₹)	Subscription price
Hosted GPU compute	5,500	2,850	Google Colab Pro: US$9.99 (≈₹950) per month × 3
Additional on-demand GPU credits	4,000	0	Colab pay-as-you-go: US$9.99 (≈₹950) per 100 units; not required
Cloud storage (≈ 200 GB)	1,500	630	Google One 200 GB: ₹210 per month × 3
Software	0	3,207	Microsoft 365 Personal ₹689 + GitHub Pro US$4 (≈₹380) per month × 3; Python, PyTorch, LibreOffice: no licence fee
Datasets and encoder weights	0	0	Open licences (OpenNeuro; TRIBE v2 CC BY-NC 4.0)
Documentation and reprographics	1,500	printing	Report binding, A3 poster
Contingency (≈ 15%)	2,000	0	Not required
Total	**14,500**	**6,687 + printing**	

Prices are current Indian list prices (US$ at ₹94.95, 23 Oct 2026). Manpower: 580 planned person-hours (330 + 250), as in Report I.

Individual contributions

Abhranil Sarkar (23BML0077)

Computational track

- Model instrumentation; the differentiable re-entry point
- Inversion engine and optimisation
- Synthetic and control experiments; verification tests
- Real-fMRI pipeline: registration, surface projection, features, evaluation
- Figures, code release, live demo, manuscript and reports

Tanisha Choudhuri (23BML0081)

Neurobiological track

- Functional neuroanatomy; six networks from the Destrieux atlas
- Curation of the LeBel dataset
- Interpretation of the regional and real-data findings
- Ethics and SDG framing
- Reports, reviews and the poster

PO and PSO mapping

Outcome	Justification from this project
PO1 Engineering knowledge	Linear algebra, optimisation and inverse-problem theory applied to a 177 M-parameter neural operator
PO2 Problem analysis	Decoding formulated as an ill-posed inverse problem; causes of incomplete recovery separated by falsifiable tests
PO3 Design of solutions	Inversion pipeline, its controls and the real-fMRI surface projection
PO4 Investigation	Controlled experiments with exact ground truth, matched random controls, verification re-run
PO5 Modern tools	PyTorch, TRIBE v2, ANTs, nilearn, GPU (Colab) computing
PO6, PO8 Society and ethics	Mental-privacy and clinical-use implications; licences and data-use terms respected
PO9, PO10 Teamwork, communication	Two-track team; IEEE manuscript, report, poster, reviews, live demo
PO11 Project management	Seven-stage schedule met within the Report I budget (₹6,687 in subscriptions + printing, against ₹14,500)
PO12 Life-long learning	Self-taught foundation models and neuroimaging tools
PSO2, PSO3	Analysis and imaging of biosignals (fMRI) with modern computational tools

WP and WK mapping (complex engineering problem)

Attribute	How the project meets it
WP1 Depth of knowledge	Neuroscience, fMRI physics, deep learning and inverse-problem theory together
WP2 Conflicting requirements	Decoding fidelity vs non-invasive, low-cost acquisition vs mental privacy
WP3 Depth of analysis	No closed-form inverse; non-convexity separated from information loss by forward-only tests
WP4 Familiarity of issues	Slow, noisy, blurred BOLD; no ground truth in real data
WP7 Interdependence	Anatomical ROI definition and computation depend on each other
WK3 / WK4	Signals and systems; inverse problems and conditioning
WK5 / WK6	Design of the inversion and projection pipeline; modern computational tools
WK8	Current research literature (2023–2026 encoders and decoders)

Social and environmental impact

Social

Concerns people who keep comprehension but lose speech; measuring what fMRI decoding can and cannot recover directs effort towards approaches that can work and away from over-promising.

Mental privacy: a near-perfect encoder identifies which known moment was heard but does not reconstruct the representation, evidence for neural-data policy (OECD 2019, UNESCO 2025). No new human data collected.

Environmental

Re-used a pretrained model instead of training a foundation encoder, avoiding its large energy cost.

All computation ran on one shared cloud GPU and CPUs for a modest number of GPU-hours; the public release lets others reproduce results without re-running everything.

Limitations

- Synthetic ground truth lives in a random low-dimensional subspace, not real language embeddings; qualitative findings are expected to transfer, absolute values may not.
- Inversion targets the aggregated representation x, one stage after raw LLaMA embeddings.
- All solutions from Adam; the SGD / L-BFGS control did not converge under the settings tried.
- No haemodynamic operator in the released model, so temporal bandwidth limits cannot be measured.
- Release ships only the shared 'average subject' layer, so per-subject analysis is not possible with public weights.
- Regional z-scores use a random-vertex null that ignores spatial autocorrelation; a spin test is stricter.
- Regional inversion uses one ground truth and one budget (8,000 steps, CPU); comparisons are internal to that run, and faster convergence cannot yet be separated from a better optimum.
- Real-data test: one subject (UTS03), one held-out story, 5 inversion queries; per-network inversion on real data needs held-out stories from more subjects.
- Corrected barrier test on four ground truths with eight starts each; original perturbation test withdrawn.

Future work

1. **Real data at scale** — Inversion across all eight LeBel subjects; per-network real-data test on genuinely held-out stories.
2. **Better search** — Optimisers and priors matched to the operator's weak directions; restarts and ensembling over minima; loss re-weighting away from high-gain, low-diversity regions.
3. **Earlier entry point** — Invert the raw extractor features and add an explicit haemodynamic model.
4. **Stricter statistics** — More ground truths and inits for the barrier test; spin-permutation nulls for regional z-scores; a converged second-order optimiser control.
5. **Other modalities** — Apply the same identifiability tests to EEG and MEG encoders as they appear.

Conclusion

A near-perfect brain encoder does not by itself make meaning recoverable from fMRI. Recoverability has to be measured, not inferred from predictivity.

- Encoding error falls 28.4× while recovery saturates at 0.944: accuracy does not certify decoding
- Equal fits disagree (0.817) and are separated by barriers (4.8×; 8.1× to a deeper minimum, 4 truths): the inverse is non-convex
- What is not recovered lies in the operator's weakest directions (2.15× below random)
- Visual cortex, the most responsive, recovers worst (−0.125); somatomotor, the least, best (+0.102)
- Real fMRI: 56% of the noise ceiling; 4 of 5 held-out segments identified among 45, yet cosine 0.02
- Everything is released: manuscript, report, poster, code, results, live demo and four notebooks

Backup · Anticipated questions

Slides held in reserve for the viva

Q: Is the non-convexity just Adam?

Couldn't Adam's implicit bias create the different answers?

The barrier is shown with forward passes only: the error along the straight line between two distinct equal fits rises 4.8× (four truths). The weak-direction result is a Jacobian probe. Neither involves an optimiser. Different initialisation distributions reach different minima reproducibly across four ground truths.

Then why did SGD and L-BFGS give different answers?

They never reached comparable fit (1.4e-4 and 2.8e-5 vs 3.5e-7), so they are not valid comparisons. We report the control as failed rather than interpret it.

What would settle it?

A tuned second-order method reaching ~1e-7 fit from the same initialisations. Listed as future work.

Q: Regional analysis details

Why Destrieux and not Glasser or Schaefer?

Destrieux ships with Nilearn on the same fsaverage5 mesh TRIBE predicts on, with standard anatomical names. Network membership comes from the language (Fedorenko) and default-mode (Buckner) literature. HCP-MMP (Glasser) is a natural robustness check.

Are the regional differences just ROI size?

No: every ROI is compared with a random vertex set of exactly the same size, and two networks have a second random seed to bound control variability.

Why no left-lateralisation of language?

The release is an average-subject model; TRIBE v2's own paper finds lateralisation in contrasts (sentences vs word lists), not in raw sensitivity. Our homologue result is consistent with bilateral responses to natural speech.

Q: The surprising regional result

How can 1,487 somatomotor vertices beat the whole cortex (0.851 vs 0.836)?

With a uniform MSE, the whole-cortex loss is dominated by high-gain, low-diversity regions such as visual cortex, which steer the search badly. Observing only diverse, low-gain vertices removes that pull. It suggests re-weighting the objective, a testable next step.

Isn't the visual result an artefact of the synthetic video dimensions?

Visual cortex is worse than its matched control on text- and audio-derived dimensions too (−0.12 and −0.14), so it is not specific to the video third of x.

Would it hold at convergence?

Unknown: all regional runs use one 8,000-step budget and one ground truth. Faster convergence and a better optimum cannot yet be separated; we claim relative differences within the run.

Q: What does 0.82 cosine mean? Is any of this real?

Is 0.82 good or bad?

It is the agreement between two equally good answers, so it bounds how reproducible a single decoded answer is. It is a property of the inverse, not a score to maximise.

Your results are all synthetic. Are they real findings?

They are exact measurements of a real, published operator, and the real-fMRI test on a held-out story shows the same pattern: identification 4/5, cosine 0.02.

Why should a clinician care?

Because the bottleneck is search, not encoder size: better decoders will come from restarts, ensembling over minima and language-derived priors.

Q: Why fMRI, not EEG? Why TRIBE v2?

Why not EEG?

Brain2Qwerty (Lévy et al., arXiv 2502.17480): 32% character error with MEG vs 67% with EEG on the same architecture.

Why not simultaneous EEG–fMRI?

The best open simultaneous dataset (NKI, 22 adults) has rest, Inscapes and short movie clips, no sustained language: data-limited, not method-limited.

Why TRIBE v2?

Largest open-weight encoder: 451.6 h training, 1,117.7 h evaluation across 720 subjects, all 20,484 cortical vertices, first at Algonauts 2025.

Q: Your barrier result changed. Why should we trust it now?

What was wrong?

The original test picked the least similar pair among all eight starts, which included an unconverged one (Gaussian s.d. 5, error 3.2× the best). Its 2.6× → 0.6× numbers are withdrawn.

What did the re-run do?

Eight starts for 14,000 steps on a T4, four ground truths; only the five blind starts within 2× were compared; the error was measured at 21 points on each straight line. Every distinct pair rises (29 of 40: 2.4–8.2×, median 4.8×); near-identical pairs share one minimum.

Does it change the conclusion?

No, it strengthens it: equal fits are separate minima, and near-truth starts reveal a deeper minimum 8.1 ± 3.2× barrier away. The selection rule is fixed in the code and the results are in the repository.

Q: Is the synthetic ground truth valid?

How was it built?

In TRIBE v2's own input space (32 × 1,152): 8 random directions with their own cosine time courses, unit s.d., four seeds; white noise as a control. We push it through the frozen model, y = f(x), and recover x from y alone.

Why is it 'correct'?

By construction: we chose x, so the recovery error is exact. It is the standard known-answer test for inverse problems and assumes nothing about the brain; the forward model is the real released one (r = 1.0000).

Is it realistic?

Real story representations have 4.9 effective dimensions (84% of variance in 8), s.d. 1.10; results do not change with white noise or a 64× change in speed; and a real 16-s representation as ground truth gives the same picture, more strongly (disagreement 0.19–0.56, barriers 11–21×).

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